

Order Cancellations, Fees, and Execution Quality in U.S. Equity Options

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Abstract

We examine the effects of an order cancellation fee on limit order flow and execution quality in the PHLX options market. The cancellation fee on professional order flow is effective in reducing the rate at which limit orders are canceled. While the cancellation fee discourages the submission of non-marketable orders, it encourages the submission of marketable orders. Consequently, non-marketable order fill rates increase, marketable order fill speeds decrease, and bid-ask spreads widen. We also find slight increases in both dollar volume and market share. We explore whether a change in pick-off risk is a potential explanation for these new findings and policy implications.

Keywords: Limit Orders, Cancellation Rates, Fees, Execution Quality

Acknowledgements: The authors would like to thank NASDAQ for providing order data and LiveVol for providing trade data across all public options exchanges. We would also like to thank Robert Battalio, Amy Edwards, Kathleen Fuller, Michael Goldstein, Thomas McInish, Bonnie Van Ness, seminar participants at the University of Mississippi, University of Memphis, West Virginia University, Utah State University, Boise State University, St. John's University, the University of Dayton, Central Michigan University, Kent State University and participants at the 2017 Financial Management Association Annual Conference.

1. Introduction

Research documents high levels of order cancellation activity in financial markets. For instance, Hasbrouck and Saar (2009) find that nearly one-third of all non-marketable limit orders for NASDAQ equity securities are canceled within two seconds of submission, and the U.S. Securities and Exchange Commission (SEC) documents that over 96 percent of orders placed in the equity markets in the second quarter of 2013 were canceled.¹ High levels of order cancellation activity are also documented in the options markets. For example, on June 5, 2013, there were over one billion quotes submitted for SPY options, and only one in 11,254 of those quotes executed.² While limit order traders might be better able to mitigate both non-execution and adverse selection risks by quickly canceling and revising their orders upon the arrival of new information (Foucault, Röell, and Sandås, 2003; Jovanovic and Menkveld, 2016), the canceled order flow might disrupt the matching process between willing buyers and sellers (Angel, 2014). Therefore, the effect of order cancellation activity on the trading welfare of market participants is indeterminate and an empirical question.

The primary purpose of this study is to examine the effects of an order cancellation fee on limit order trading and order execution quality in equity options markets. We utilize the introduction of an order cancellation fee on the NASDAQ OMX Philadelphia Options Exchange (PHLX), and the absence of any similar cancellation fee on the NASDAQ Options Market (NOM), as a natural experiment. On August 19, 2010, the PHLX filed a proposal with the SEC to assess separately a \$2.10 cancellation fee on customer orders and a \$1.10 cancellation fee on professional all-or-none (AON) orders. The exchange anticipates that the removal of these order cancellations will improve the trading process for all market participants (see SEC Release No. 34-62744). To the extent the cancellation fee improves order execution quality, then competing options exchanges with similar market structures may consider enforcing a similar fee. However, if the cancellation fee deteriorates order execution quality, then competing options exchanges may be discouraged from adopting a similar fee.³

¹ See “Trade to Order Volume Ratios” market structure research from the U.S. SEC released on October 9, 2013.

² See the research analysis posted by Nanex, LLC at <http://www.nanex.net/aqck2/4308.html>.

³ As trading in options is shown to contribute to price discovery in the underlying equity markets (see e.g. Manaster and Rendleman, 1982; Easley, O’Hara, and Srinivas, 1998; Chakravarty, Gulen, and Mayhew, 2004), the results of this paper may also apply to equities.

Submitting orders and quickly canceling those orders is common and frequent in current financial markets (see e.g. Ellul, Holden, Jain, and Jennings, 2007; Hasbrouck and Saar, 2009; Van Ness, Van Ness, and Watson, 2015). Liu (2009) and Goldstein, Kumar, and Graves (2014) contend that advancements in technology and the transition of exchanges to electronic trading platforms are possible explanations for the high cancellation rates observed in financial markets. High cancellation activity might also simply be the result of liquidity providers canceling and revising their bid and ask quotes as they closely monitor the news and other quote updates (Foucault, Roell, and Sandas, 2003). Malinova, Park, and Riordan (2016) note that a trade involves four separate order messages from the market maker: one for the trade, one for the cancellation on the opposite side of the book, and two for the revised bid and ask quotes. The effect multiplies when market makers are quoting at more than one trading venue contemporaneously. During our sample period, there are eight public equity options exchanges; therefore, a single trade could naturally result in eight cancellation messages. As limit orders are canceled and resubmitted in response to shifts in supply and demand, there is no reason to believe that this strategy is harmful to order execution quality.

However, there are some potential nefarious explanations for the high levels of order cancellation activity observed in financial markets. For example, in December of 2015, the SEC announced fraud charges against three Chicago-based traders using hidden AON orders to circumvent rules in options markets. The traders would submit hidden AON orders for a particular option series, and simultaneously place smaller non-bona fide displayed orders for the same option series on the opposite side of the order book. The displayed orders would push the best bid and offer prices towards the AON orders, which would ultimately execute. The traders would then cancel the displayed orders. The NASDAQ also fined Citadel Securities LLC (CDRG) for sending millions of orders to the exchange with few to no executions (see NASDAQ No. 20100223345-02). For instance, on August 25, 2011, in one second, CDRG sent 13,917 orders for 2,410,700 shares, all of which were displayed and canceled before execution. If displayed limit orders are canceled and not resubmitted, then this can decrease the supply of liquidity and negatively affect order execution quality. Market participants who seek to interact with orders that are canceled before their order arrives may achieve less favorable executions, or no execution at all (see SEC Release No. 34-65610).

The execution quality of limit orders may depend on the trade-off between non-execution risk and adverse selection risk (Sandås, 2001; Colliard and Foucault, 2012). An order cancellation fee might improve execution quality if it increases the probability (and/or speed) of completing a trade. Since limit orders are stored in the order book and do not demand immediacy, the execution of a limit order is not guaranteed. Colliard and Foucault (2012) argue that trading fees affect both the execution probability for limit orders and the ratio of market to limit orders. We contend that an order cancellation fee may increase monitoring costs for limit order traders, as it is more expensive to withdraw and revise non-marketable order submissions. As a result, traders may shift to marketable orders in an attempt to pick off the exposed non-marketable limit orders. Copeland and Galai (1983) model non-marketable limit orders as free straddle options to future traders with access to new information (public or private), and Foucault, Röell, and Sandås (2003) argue that traders intensely monitor markets to detect, and pick off, these stale quotes. A shift in order flow from non-marketable to marketable may quicken the speed, and rate, with which limit orders execute (Battalio, Corwin, and Jennings, 2016).

An order cancellation fee might negatively affect execution quality if it reduces non-marketable order flow and increases the cost to trade. Limit order traders must continuously update and revise their orders, which requires close monitoring of their inventory positions. Goettler, Parlour, and Rajan (2005) develop a discrete time model of a pure limit order market and introduce a piecewise-linear cancellation function. In this model, traders cancel their orders if the consensus value moves away from their limit order price, which allows them to submit orders closer to their reservation values. Baruch and Glosten (2013) argue that traders manage the risk of getting undercut while sitting on the limit order book by quickly adjusting their limit orders. Imposing a fee on order cancellations might expose displayed limit orders to greater pick-off risk as it is more expensive to withdraw or revise order submissions. Consequently, traders might submit fewer non-marketable limit orders, which may widen bid-ask spreads.

To examine our research questions, we obtain order-level data on two options exchanges, the PHLX and the NOM, and trade-level data across all eight options exchanges. To conduct difference-in-difference (DID) tests, we match option series across exchanges. We begin our analysis by examining whether the order cancellation fee is effective in discouraging traders from canceling their displayed limit orders. This is to be expected, as the fee change places a direct cost on canceled order flow. In our DID regressions, we find that the average order cancellation rate on

the PHLX, relative to the NOM, declines by 11.97 percentage points after the fee change. We also find a decrease in the number of time-weighted runs in cancellation messages on the PHLX over 10-minute intervals during the day (see Table A.1 in the Internet Appendix). Furthermore, we find that, relative to limit orders submitted to the NOM, limit orders submitted to the PHLX remain on the book for a longer period of time before they are canceled after the fee is enforced. Since the limit orders submitted to the PHLX appear to be less fleeting after the fee change, the orders potentially reflect more committed trading sentiment (Angel, 2014; Friederich and Payne, 2015).

Next, we examine whether the order cancellation fee affects order routing decisions. We find that the average number of non-marketable orders submitted to the PHLX, relative to the NOM, decreases by 82.52% after the fee change. This decline appears to be primarily driven by the departure of professional traders supplying liquidity, as the percentage of non-marketable orders represented by professionals on the PHLX decreases from 42.9% in the pre-fee period to 25.16% in the post-fee period. In contrast, the average number of marketable orders on the PHLX, relative to the NOM, increases by 15.82% after the fee is introduced. Therefore, it seems that market participants submit more marketable orders, and less non-marketable orders, to the PHLX after the fee change.

Because the order cancellation fee appears to impact traders' order routing decisions, it might also affect the execution quality of their orders. We address this question by analyzing three measures of order execution quality: the likelihood of a fill, the speed of a fill, and the effective bid-ask spread at the time of the fill. Consistent with Harris and Hasbrouck (1996), we find that an increase in marketable order flow is associated with higher order fill rates. Specifically, the average non-marketable order fill rate on the PHLX, relative to the NOM, increases by 7.3 percentage points after the fee change. We also find that the order cancellation fee is associated with quicker marketable order fills. Since both the probability of execution and speed of execution are important to traders (Colliard and Foucault, 2012; Garvey and Wu, 2009), our results indicate an improvement in at least two aspects of execution quality. However, we also find a 14.93 percent increase in the average effective spread on the PHLX, relative to competing options exchanges, after the fee change.

Since there are costs and benefits associated with the order cancellation fee, we also examine whether overall trading activity on the PHLX changes as participants adjust to the new trading environment. For all trades, the average market share on the PHLX increases by 0.45

percentage points after the cancellation fee is introduced. However, for trades when all venues are at the best quotes, the average market share on the PHLX increases by 2.79 percentage points after the fee change. Additionally, we find that the average dollar volume on the PHLX, relative to competing options exchanges, increases by 8.14% after the fee is enforced (see Table B.1 in the Internet Appendix). Even if market share and dollar volume were to remain the same around the fee change, our results suggest that the PHLX was able to reduce cancellation activity without losing trading activity.

In our final set of tests, we examine whether pick-off risk increases around the introduction of the order cancellation fee, which can possibly explain the change in order routing behavior and execution quality observed on the PHLX. We proxy pick-off risk using the price impact of limit order fills and two measures of volatility. We find that the average price impact increases more for limit orders filled on the PHLX than those on the NOM after the fee change. We also find that both midpoint volatility and range volatility are higher on the PHLX, relative to the NOM, after the fee change. Combined with our previous findings, and consistent with the arguments of Hasbrouck and Saar (2001), the increase in pick-off risk seems to reduce the desirability of non-marketable orders and increase the desirability of marketable orders. While the reduction in non-marketable orders appears to widen bid-ask spreads, the influx in marketable orders seems to increase order fill rates. Thus, consistent with Liu (2009), it appears that limit order traders, when facing monitoring constraints, such as an order cancellation fee, must weigh the risk of non-execution with that of pick-off risk.

Our article contributes to a growing literature that examines order cancellation activity in financial markets (see e.g. Boehmer, Saar, and Yu, 2005; Hasbrouck and Saar, 2009; Van Ness, Van Ness, and Watson, 2015). Our study also has important policy implications, as the issue of traders who cancel a lot of their orders has received significant attention and debate from policy makers, regulatory agencies, and exchange officials. For instance, former U.S. Democratic presidential candidate Hillary Clinton proposed a tax on high-frequency trading (HFT), targeting securities transactions with “excessive” levels of order cancellations, under the presumption that such trading strategies are abusive and detrimental to financial markets.^{4,5} In response to the flash

⁴ The HFT-specific aspects of the broad proposals for the financial system provided by Hillary Clinton in an op-ed piece in *The New York Times* on December 7, 2015 entitled, “How I’d Rein in Wall Street.”

⁵ See also a September 7, 2010 address entitled, “Strengthening Our Equity Market Structure,” issued by former SEC Chairwoman Mary L. Schapiro.

crash on May 6, 2010, the Commodity Futures Trading Commission (CFTC) and the U.S. SEC recommended a uniform fee across all exchanges, to fairly allocate the costs imposed by high levels of order cancellation activity.⁶ Exchange officials at the NASDAQ proposed a minimum-life order type on the PSX, with the intent on encouraging longer-lived limit orders (Jones, 2013). The results in this study add to our understanding of how a cancellation fee on limit orders affects both order flow and execution quality.

2. The PHLX Cancellation Fee Policy

Exchange officials on the PHLX acknowledge the costs associated with high levels of order cancellations and, subsequently, have enforced a fee to help monitor message-intensive trading strategies. The purpose and statutory section of the rule filing (see SEC Release No. 34-62744, page 2) states:

“The Exchange has observed that the number of canceled professional AON orders greatly exceeds the normal order cancellation activity on the Exchange for all other order types, and thus affects the automated order handling capacity of the Exchange’s systems... The Exchange believes that the proposed amendments are reasonable because they will ease system congestion and allow the Exchange to recover costs associated with excessive order cancellation activity.”

Effective August 19, 2010, the PHLX began to assess separately, a \$2.10 order cancellation fee on customer limit orders and a \$1.10 per order fee on canceled AON professional orders.⁷ The PHLX amended the fee policy to add a cancellation fee on professional AON orders to the existing cancellation fee on customer orders. The exchange believes that separately calculating professional and customer cancellations will simplify the calculation of the fee and reduce system congestion on the exchange. The term professional refers to any person or entity that (1) is not a broker or a dealer, and (2) submits more than 390 orders in listed options per day on average during a calendar month. An order cancellation fee is only assessed in a month in which more than 500 electronically delivered orders are submitted and canceled by the same member.⁸

⁶ See page 11 of the summary report of the joint CFTC-SEC Advisory Committee on Emerging Regulatory Issues “Recommendations Regarding Regulatory Responses to the Market Events of May 6, 2010.” See also page 47 of the January 14, 2010 SEC CFTC Concept Release on Equity Market Structure and *The Wall Street Journal* article on February 23, 2012, “SEC May Ticket Speeding Traders: High-Frequency Firms Face Fees on Canceled Transactions.”

⁷ See the NASDAQ Options Trader Alert #2010 – 53 for a detailed description of the updates to the cancellation fee assessment criteria effective August 19, 2010. See also the SEC Release Nos. 34-62744 and SEC 34-62745 for the notice of filing and immediate effectiveness of the proposed rule change relating to the cancellation fee.

⁸ Electronic orders are delivered through the Exchange’s options trading platform.

AON limit orders are orders that must execute in their entirety or not at all. The fee applies to AON orders that are deliberately canceled, rather than those that cancel by order design. Traders are not required to show an AON order to the public (i.e. hidden). Therefore, the AON order price may not be reflected in the marketplace, preventing any partial execution.⁹ For this reason, we focus on complete order fills throughout the analysis. We also note that only customers and professionals may submit AON orders to the PHLX market.

The hidden feature of AON orders might encourage market participants to supply liquidity who otherwise would not. Tuttle (2003) shows that hidden orders on the NASDAQ adds “substantial” liquidity to the marketplace. In fact, Tuttle shows that for the NASDAQ 100 in 2001 to 2002, non-displayed orders represented 25% of dollar depth at the NBBO. Anand and Weaver (2004) argue that hidden liquidity represents depth that would otherwise not be exposed. In a study of the Euronext market, D’Hondt, De Winne, and Francois-Heude (2003) provide evidence that hidden order traders are primarily suppliers of liquidity who focus on the few ticks surrounding the BBO in the limit order book. However, unlike a hidden order that reveals its remaining shares after a partial execution, an AON order must execute in full, which never allows for any remaining unexecuted shares to be picked off by a third party trader with more, or better information. Therefore, a liquidity provider might use an AON order to avoid being picked off as market conditions change. The cancellation fee on AON orders may discourage professional traders from submitting liquidity supplying orders as those orders would be more exposed to pick-off risk. This reduction in liquidity provision could lead to an increase in the cost to trade.

Bessembinder, Panayides, and Venkataraman (2009) contend that hidden orders are less likely to fill and take longer to execute. Anand and Weaver (2004) examine the decision of the Toronto Stock Exchange to ban and then reinforce hidden orders. The authors find that if traders are required to expose their orders they will not leave the market, but rather they will submit more marketable orders. Therefore, the cancellation fee may encourage the use of other order types, such as marketable orders, that are more likely to fill and at a faster pace. Direct access users and other professional traders have the ability to attach various conditions to their orders, such as price specifications, routing procedures, and expiration times, which can slow down execution speed. Marketable orders should, on average, have fewer of these conditions relative to non-marketable orders (see Garvey and Wu, 2009).

⁹ See discussion of AON orders at <https://www.interactivebrokers.com/en/index.php?f=577>.

3. Data Description

We gather data from several sources. The NASDAQ OMX PHLX market data feed provides the current state of simple and complex orders on the PHLX book. This includes nanosecond information on orders added and changes made to orders on option series on the PHLX limit order book. While an option class refers to a unique underlying stock, the more refined option series is defined as a particular underlying stock, call or put, strike price, and expiration date. The PHLX reports a simple order message when a single order is received or any change is made to an order. “Simple order” messages include the following fields: nanosecond time stamp, day-unique order id, market side (buy or sell), underlying security symbol, expiration date, explicit strike price, option type (call or put), original order volume, executable order volume (can increase or decrease as the size available for trading changes due to away exchange routing), order status (open, filled, or canceled), limit price, time in force (day order or good till canceled), and customer/firm identifier (customer, firm, market maker, broker/dealer, or professional). We eliminate orders reported before 9:45 a.m. and after 3:50 p.m. from our sample because the opening and closing rotations impede option series from trading freely. Complex orders, such as spreads and straddles, are priced as a package, so we also remove them from the sample.

To conduct our DID analysis, we obtain historical order-level data on a competing U.S. options exchange, the NOM.¹⁰ The NASDAQ Historical TotalView-ITCH to Trade Options (ITTO) is a direct data feed that provides millisecond view of simple equity options on the NOM. This includes orders added and changes made to orders resting on the NOM limit order book. We download several different types of messages that are linked by a unique order reference number. “Add order” messages are time stamped records for new orders added to the book, including order time, market side (buy or sell), order size (# of contracts), option type (call or put), expiration date, explicit strike price, and order price. “Executed order” modification messages are time stamped records generated by (partial) executions and report executed contracts and execution price (when the execution price differs from the add order price). “Order cancel” messages are time stamped records generated by partial cancellations and report the number of contracts canceled. “Replace”

¹⁰ For each of our different outcome variables we test the parallel trends assumption and find insignificant differences between the treatment group and the control group(s) in the pre-cancellation fee period. Inferences can be made from the figures discussed in the results sections.

messages are time stamped records that report the new order reference number, new order price, and new order size. “Delete” messages are time stamped records that report when an order is deleted from the NOM order book.

We acknowledge that the NOM is not the ideal control exchange. While the PHLX is a hybrid floor and electronic market with both quotations and orders, the NOM is a purely electronic order-driven market. Due to data limitations we are unable to obtain order-level data for other competing exchanges and, therefore, we are restricted to the PHLX and the NOM in our order-level analysis. However, to control for potential heterogeneity in the trading of options between the PHLX and the NOM, we perform a daily match across option series. We also obtain price and volume information on trades and on current bids and offers across all eight equity options exchanges (NOM, AMEX, CBOE, ISE NYSE ARCA, PHLX, BOX, and BATS) from a data technology company LiveVol. Options exchanges report both trade and quote data to the Options Price Reporting Authority (OPRA) and LiveVol creates historical trade files that include trade price, trade size, an exchange identifier where the trade prints, the best bids and offers (BBO) across each exchange at the time of the trade, the National Best Bid and Offer (NBBO), the underlying bid and ask prices, implied volatility, and calculated delta. We follow Battalio, Hatch, and Jennings (2004) and combine multiple executions in the same option series, executing on the same exchange at the same price at the same time with the same trade condition identifier into a single trade. We also eliminate BBOs that are either locked or crossed (see Battalio, Shkilko, and Van Ness, 2016).

[Insert Table 1 Here]

Our sample period ranges from July 26, 2010, which is the first day that order-level data is available on the PHLX, to October 18, 2010. Due to data corruption issues in the OPRA feed, we drop August 13th and September 2nd. Therefore, we are left with the 17 days prior, and the 40 days following, the fee change on August 19th, 2010. The initial sample consists of 143,670 unique option series (underlying asset, call or put, expiration date, and strike price). To concentrate on the most active option series, we exclude series with fewer than five orders in a particular day, one of which is required to fill. This reduces the sample to 15,569 unique option series. To ensure accurate comparisons across exchanges, we conduct a daily match between options series originating on the PHLX with those originating on the NOM. This reduces our sample to 8,844 option series. Finally, we merge the order data with closing prices and shares outstanding from the Center for

Research in Security Prices (CRSP), and retain options on common stocks and ETFs. Our final sample consists of 7,492 unique option series on 116 underlying securities.¹¹ We focus the majority of the analysis on the 5,850 option series on common stocks, but we find that our results are similar for the 1,642 ETFs. In unreported results, we find that these option series account for just under 50% of total order volume on the PHLX during the sample period.

4. Empirical Results

4.1. Order Flow

We begin our empirical analysis by investigating the effect of the PHLX cancellation fee on order routing decisions. We separate orders on the PHLX and NOM into marketable and non-marketable. We do not observe quotes in our dataset, therefore, we approximate a marketable order as a limit order that fills within 500 milliseconds of submission.¹² We believe it is important to distinguish between marketable and non-marketable orders, as limit order traders submitting marketable orders may be more concerned with the risk in the time-to-execution, while those placing more passive non-marketable orders may be more concerned with the risk of non-execution. Therefore, order submission strategies between marketable and non-marketable limit order traders might differ around the enforcement of the cancellation fee.

[Insert Figure 1 Here]

First, we examine whether or not the PHLX order cancellation fee deters traders from canceling their limit orders. We measure order cancellation rates on a daily basis by option series as the number of orders canceled divided by the total number of orders added to the book. Panel A of Figure 1 provides a visual representation of how the order cancellation fee on the PHLX impacts cancellation activity. We plot average order cancellation rates on the PHLX (dark solid line) and the NOM (light dotted line) in event time. The vertical dotted line represents the effective date of the cancellation fee. We show that order cancellation rates for options on the PHLX decline substantially around the fee change, and remain at a lower rate in the two months following the effective date. In contrast, order cancellation rates for options on the NOM remain above 99

¹¹ The results are robust to the relaxation in the order/fill screening criteria.

¹² The results are robust to different millisecond cutoff levels 50, 100, 200, and 1,000. E*TRADE states that the average time to execution on marketable limit order for a non-S&P 500 security is 180 milliseconds (see <https://us.etrade.com/trade/execuion-quality>).

percent over the entire sample period. Thus, the order cancellation fee appears effective in reducing order canceling activity on the PHLX.

Since the majority of fee changes on an exchange are made on monthly intervals, it is a rare occurrence for a fee change to publish and become effective mid-month. The data seems to suggest that the “true” effective date was closer to the end of August, 2010. It could be that firms simply assumed that the change would go into effect the following month, similar to other fee changes. Alternatively, the exchange calculates the 500 order threshold in a particular calendar month and then assesses the per order fee at the end of the month. Therefore, the fees for August would not be calculated until the end of the month, which could have delayed the reaction of traders to the new pricing policy.

Our results show that the cancellation fee has an immediate impact on order cancellation activity. However, an important concern is whether or not the cancellation fee had a long-term impact on order routing behavior. In Panel B of Figure 1, we plot average daily order cancellation rates across option series on the PHLX in the month of September for the period 2010 to 2016. We observe a substantial decline in order cancellation rates on the PHLX after the fee is introduced, which remains low for the following six years. This longitudinal analysis suggests that the cancellation fee affects order routing decisions long-term.

Another concern is how order cancellation rates on the PHLX compare to order cancellation rates on other competing exchanges today. We are able to obtain order level data that allows us to calculate cancellation rates on four U.S. options exchanges for the month of September in 2016: PHLX, NOM, Bats BZX, and Bats EDGX. We find that the average order cancellation rate on the PHLX, 62%, is substantially lower than on the other three options exchanges. In fact, the two Bats exchanges have average daily order cancellation rates higher than 97%, while the NOM has an average daily order cancellation rate of 90%. Our results seem to suggest that the cancellation fee on the PHLX discourages limit order traders from canceling their orders.

Next, we examine if the PHLX order cancellation fee encourages longer-lived limit orders. An important decision traders make each and every time they submit a limit order, is how long they allow that order to remain on the order book. Similar to Hasbrouck and Saar (2001) and Lo, MacKinlay, and Zhang (2002), we estimate the duration between when an order is submitted and subsequently deleted. Market participants complain about “phantom liquidity,” where orders disappear when they attempt to trade against them (Angel, 2014). Traders may be unable to

distinguish between a firm quote that can be traded upon and a fleeting quote that is canceled within nanoseconds of submission. An order cancellation fee should discourage traders from submitting frivolous orders and, therefore, increase the duration between order submission and cancellation. Panel A of Figure 2 plots order duration, in seconds, over the sample period. The dark solid (light dotted) line represents order duration on the PHLX (NOM). While the average daily order duration on the PHLX increases substantially around the enforcement of the cancellation fee, the average daily order duration on the NOM remains constant. In fact, the average time between submission and cancellation in the post-fee period is approximately 400 seconds longer for orders submitted to the PHLX, relative to the pre-fee period.

[Insert Figure 2 Here]

We also examine if the order cancellation fee discourages the submission of non-marketable limit orders, while encouraging the submission of marketable limit orders. Since the NOM scale of non-marketable and marketable orders is much larger than on the PHLX, we take the natural log of both to make more accurate comparisons across the sample period. Panel B of Figure 2 shows that the average number of non-marketable orders submitted to the PHLX decreases following the fee change. We do not, however, find a similar downward trend in the average number of non-marketable orders submitted to the NOM. We interpret this as evidence of a decrease in the liquidity provision around the introduction of the PHLX order cancellation fee as it becomes more expensive for limit order traders to revise their quotes. In unreported tests, we find that quoted depth declines around the introduction of the order cancellation fee. This is consistent with Friederich and Payne (2015), who show that after the introduction of the fee in Italy for order-to-trade ratios in excess of 100:1, quoted depth declines. Panel C of Figure 2 shows a gradual increase in marketable order flow on the PHLX around the fee change. While the number of marketable orders on the NOM is fairly volatile over the sample period, we do not find a directional change. We note that the average number of marketable orders submitted to the PHLX is noticeably less than the NOM in the pre-fee period, which becomes nearly indistinguishable in the post-fee period.

To control for other factors that might influence order routing decisions, we estimate specifications of the following regression equation on average daily option series observations:

$$\begin{aligned}
Order\ Flow_{i,j,t} = & \alpha + \beta_1 Phlx_j + \beta_2 Post_t + \beta_3 Phlx \times Post_{j,t} \\
& + \beta_4 Days - to - Expiration_{i,j,t} + \beta_5 S/X_{i,j,t} + \beta_6 Order\ Price_{i,j,t} \\
& + \beta_7 Fragmentation_{i,j,t} + \beta_8 Underlying\ MCAP_{i,t} \\
& + \beta_9 Log(Underlying\ Volume)_{i,t} + \delta_t + \gamma_i + \varepsilon_{i,j,t},
\end{aligned} \tag{1}$$

where the dependent variable is set to one of three order flow measures: *order cancellation rate*, *order duration*, *log (# of non-marketable orders)*, or *log (# of marketable orders)*. Each variable is calculated on a daily basis by option series. *Phlx* is a categorical variable equal to one if the order is submitted to the PHLX and zero if the order is submitted to the NOM. *Post* is a dummy variable equal to unity if the order is submitted in the post-fee period and zero otherwise. The interaction term between *Phlx* and *Post* is the DID estimator. *Days-to-Expiration* is the number of trading days until option expiration. *S/X* is the ratio of the underlying stock price to the option strike price. *Order Price* is the average daily order price. *Fragmentation* is measured as the volume executed outside of the trade exchange (see O'Hara and Ye, 2011). *Underlying MCAP* is the market capitalization (price time shares outstanding) of the underlying security. *Underlying Volume* is the average daily share volume in the underlying security. In the full model specification, we include option fixed effects and day fixed effect to control for unobservable option characteristics and time trends. We do not include a *Post* dummy variable when employing day fixed effects, as this would violate the full column rank assumption for consistent estimation. We estimate variance inflation factors (VIF) for each model specification and find that all values fall below three, providing no indication of multi-collinearity. We report the results of estimating equation (1) in Table 2, with *t*-statistics in parentheses obtained from standard errors two-way clustered by option series and day.

[Insert Table 2 Here]

The first two columns of Table 2 show that the average order cancellation rate on the PHLX, relative to the NOM, significantly decreases following the cancellation fee change. The coefficient on the interaction term in the second column is -0.1197 and significant at the 0.01 level. This suggests that, controlling for option- and firm-specific factors, the average daily order cancellation rate on the PHLX decreases by 11.97 percentage points, relative to the NOM, after the fee change. Thus, the order cancellation fee appears to be effective in reducing the number of non-marketable limit orders that are submitted and subsequently deleted.

The third and fourth columns in Table 2 show that the average order duration significantly lengthens on the PHLX, relative to the NOM, after the enforcement of the order cancellation fee. The estimated coefficient on the interaction term in the fourth column is 218.5647, which is significantly different from zero at the 0.01 level. Therefore, the average time between order submission and cancellation increases by approximately 218 seconds on the PHLX, relative to the NOM, after the cancellation fee is implemented. One interpretation of this increase in order duration might be that the non-marketable orders submitted to the PHLX in the post-fee trading environment represent more committed trading sentiment (Angel, 2014).

The results in the last four columns of Table 2 can be interpreted as percentage changes in non-marketable/marketable order flow from before to after the cancellation fee is introduced, as we are estimating a log-level model. The coefficient on the interaction term in the sixth column is -0.8252 and significant at the 0.01 level. This indicates that the percentage of non-marketable orders on the PHLX, relative to the NOM, decreases by 82.52% following the enforcement of the order cancellation fee, other factors held constant. In the final column, we show that the percentage of marketable orders on the PHLX, relative to the NOM, increases by 15.82% after the order cancellation fee is introduced, other option and firm factors held constant. Our results seem to suggest that traders on the PHLX switch from non-marketable orders to marketable orders after the fee change.

To examine the change in order flow further, we separate the sample by trader type. The PHLX feed allows us to distinguish between professional, broker-dealer, customer, firm, and market maker orders. We are not able to separate order flow by trader type on the NOM. Therefore, the analysis reported in Table 3 concentrates on PHLX order flow only. We analyze the percentage of total orders, the percentage of non-marketable orders, and the percentage of marketable orders accounted for by each trader type in a particular option series before and after the introduction of the cancellation fee. We also estimate average order cancellation rates and non-marketable order fill rates by trader type.

[Insert Table 3 Here]

Panel A of Table 3 reports professional order flow on the PHLX. We document that professional orders account for 42.38% (42.90%) of all (non-marketable) orders submitted in active option series on the PHLX before the fee change and only 24.44% (25.16%) of all (non-marketable) orders after the fee change. These decreases are both statistically significant and

economically meaningful. We interpret the decline in non-marketable order flow by professional traders as evidence of many of these traders withdrawing from the PHLX market altogether – at least on the supply side. Since professional traders supplied a substantial amount of liquidity prior to the fee change, their withdrawal may have consequences for the entire PHLX market.

The two trader types that dominate the PHLX order market are customers and professionals. In fact, prior to the fee change, customers and professionals represent 92.26% of non-marketable orders, which increases only slightly to 93.44% after the fee is introduced. Since the percentage of orders across trader types must sum to one, it is not surprising that we find an increase in the percent of customer orders around the fee change. Specifically, In Panel C of Table 3, we find that the percent of non-marketable orders represented by customers increases from 49.36% in the pre-fee period to 68.28% in the post-fee period. Therefore, customer traders are forced to play a bigger role as liquidity providers after the cancellation fee is introduced.

The results in Table 3 also show that the average order cancellation rate on professional non-marketable orders decreases by 1.84 percentage points from the pre-fee period to the post-fee period. Since professional orders represented over 42% of non-marketable order flow in the pre-fee period and 98.44% of those orders were canceled, the removal of these fleeting orders might change the order strategies of other trader types. For instance, Panel B shows that broker-dealer non-marketable order cancellation rates decrease by 9.73 percentage points around the fee change and Panel D shows that firm non-marketable order cancellation rates decrease by 7.10 percentage points. Therefore, it appears that broker-dealers and firms do not have to revise their order submissions as frequently in the post-fee trading environment as the order book is potentially more static. In turn, this increases non-marketable order fill rates for professionals (1 percentage point), broker-dealers (7.98 percentage points), and firms (6.16 percentage points). We note that the average customer non-marketable order fill rate decreases by 1.38 percentage points around the fee change. However, because customers represent 68.28% of non-marketable orders in the post-fee period, and 37.54% of those orders fill, the likelihood of any given order executing appears higher in the post-fee trading environment. Thus, our results suggest that the order cancellation fee reduces professional non-marketable order flow that is canceled at an extremely high rate, which increases the probability of a non-marketable order filling.

4.2. Order Execution Quality

Since the order cancellation fee on the PHLX appears to impact the order submission process of all trader types, it might also affect the execution quality of orders. We address this question using three measures of execution quality: the likelihood of a fill, the speed of a fill, and the effective bid-ask spread at the time of the fill. To measure the likelihood of an order fill and the speed of fills, we use order-level data on the PHLX and NOM. Similar to Foucault (1999), we estimate the likelihood of a complete fill using daily non-marketable order fill rates, or the ratio of the number of non-marketable orders filled to the total number of non-marketable orders submitted (note, we do not observe each execution in the PHLX historical order files, but rather when the order is submitted and completely filled or canceled). We estimate the time-to-complete fill for both marketable and non-marketable orders as the number of seconds between order submission and complete fill. To measure bid-ask spreads, we use trade and quote data across all eight equity options exchanges. We follow Battalio, Shkilko, and Van Ness (2016) and measure dollar effective spreads as twice the absolute difference between the trade price and the exchange best bid and offer midpoint prevailing at the time of the trade (note, quoted bid-ask spreads produce similar results). To remove the influence of outliers, we winsorize effective bid-ask spreads at the 1st and 99th percentile levels.

[Insert Figure 3 Here]

In Figure 3, we plot average daily order execution quality measures for options on the PHLX in the dark solid line and the control group (NOM or Market) in the light dotted line over the sample period. In Panel A of Figure 3, we show that average order fill rates on the PHLX increase around the introduction of the order cancellation fee, and remain elevated. We do not see a noticeable change in average order fill rates on the NOM over the same time period. Therefore, it seems that the cancellation fee has a positive effect on order fill rates.

In Panel B of Figure 3, we show that average fill times for marketable orders submitted to the PHLX decreases after the fee change. We do not find a similar decline in average fill times for marketable orders submitted to the NOM. In Panel C of Figure 3, we show that average fill times for non-marketable orders submitted to both the PHLX and NOM follow a similar pattern around the event date. Therefore, in terms of order fill speed, the order cancellation fee seems to benefit aggressive liquidity demanders, but has no clear effect on the more passive liquidity suppliers.

In Panel D of Figure 3, we show that average dollar effective spreads decrease initially on both the PHLX and the other seven options exchanges, but then begin to increase in the 40 days

following the cancellation fee change. The average dollar effective spreads appear to increase at a faster rate on the PHLX in the post-fee trading environment, relative to the other seven options exchanges, as the gap between the two moving averages narrows. Since bid-ask spreads in options are extremely cyclical around expiration dates, to tease out the marginal effect of the cancellation fee it is important to control for time-to-expiration in a multivariate setting.

To more accurately control for factors that might influence order execution quality, we estimate a series of regressions. We contend that the relevant controls are option and stock attributes, order characteristics, venue traits, and time trends (see Battalio, Corwin, and Jennings, 2015; Battalio, Shkilko, and Van Ness, 2016). The unit of measurement is option series/day, and the general specification for our models is outlined as follows:

$$\begin{aligned}
 \text{Order Execution Quality}_{i,j,t} &= \alpha + \beta_1 \text{Phlx}_j + \beta_2 \text{Post}_t + \beta_3 \text{Phlx} \times \text{Post}_{j,t} \\
 &+ \beta_4 \text{Days} - \text{to} - \text{Expiration}_{i,j,t} + \beta_5 S/X_{i,j,t} + \beta_6 \text{Price}_{i,j,t} \\
 &+ \beta_7 \text{Fragmentation}_{i,j,t} + \beta_8 \text{Underlying MCAP}_{i,t} \\
 &+ \beta_9 \text{Log}(\text{Underlying Volume})_{i,t} + \delta_t + \gamma_i + \varepsilon_{i,j,t},
 \end{aligned} \tag{2}$$

where the dependent variable is set to one of three measures of order execution quality: *order fill rate*, *order fill speed*, or *dollar effective spread*. When the dependent variable is either *order fill rate* or *order fill speed*, we compare orders submitted to the PHLX versus those submitted to the NOM. When the dependent variable is *dollar effective spread*, we compare quotes and trades on the PHLX versus those on the other seven options exchanges. *Phlx* is a categorical variable equal to one if the order is submitted to the PHLX and zero otherwise. *Post* is a dummy variable equal to one if the order is submitted in the post-fee period and zero for the pre-fee period. The interaction term between *Phlx* and *Post* is the DID estimator. The remaining independent variables have previously been defined. We estimate variance inflation factors (VIF) for each model specification and find that all values fall below three, providing no indication of multi-collinearity. In the full model specification, we include option fixed effects and day fixed effects. The results of estimating equation (2) are reported in Table 4, with *t*-statistics in parentheses obtained from standard errors two-way clustered by option series and day.

[Insert Table 4 Here]

The results in the first two columns of Table 4 show an increase in the average order fill rate on the PHLX, relative to the NOM, after the introduction of the order cancellation fee.

Specifically, the average daily non-marketable order fill rate for options on the PHLX, relative to the NOM, increases by 7.3 percentage points after the fee change, other factors held constant. Combined with our results in the previous subsection, the increase in the average non-marketable order fill rate on the PHLX seems to be driven by both a decline in non-marketable order flow and an increase in marketable order flow.

The results in the third and fourth columns of Table 4 show a significant decline in the average time-to-complete fill for marketable orders submitted to the PHLX, relative to those submitted to the NOM, after the imposition of the cancellation fee. In fact, in the fourth column of Table 4, the coefficient on the interaction term suggests that the average time-to-complete fill decreases by 9.5 milliseconds for marketable orders submitted to the PHLX, relative to those submitted to the NOM, after the fee change. Therefore, the more aggressive liquidity demanders seem to be better off after the fee change. In the fifth and sixth columns of Table 4, we do not find a significant change in the average non-marketable order fill speed on the PHLX, relative to the NOM, after the initiation of the order cancellation fee.

The final two columns of Table 4 show that transactions costs increase on the PHLX, relative to the other seven options exchanges, after the order cancellation fee is imposed on professional traders. The results in the fifth (sixth) column show that the average dollar effective spread for an option on the PHLX widens by \$0.0072 (\$0.0079), relative to the other options exchanges, after the fee change. This represents a 13.61% (14.93%) increase above pre-event levels on the PHLX. Our results are consistent with studies in the equities markets, such as Malinova, Park, and Riordan (2016), who show that a tax on high-frequency market making in equities increases market-wide bid-ask spreads by 13%.

Overall, the results in this subsection seem to suggest that an order cancellation fee affects various aspects of order execution quality differently. While the average order fill rate increases and the speed of execution on marketable orders decreases, transactions costs increase. Therefore, there appears to be both advantages and disadvantages of assessing a fee on order cancellations in options markets.

4.3. Market Share

An order cancellation fee might also affect the exchange's market share as participants adjust to the new trading environment. Colliard and Hoffman (2013, 2017) show that a financial

transaction tax can reduce market participation as the cost to trade becomes more expensive. However, an order cancellation fee does not necessarily increase the cost to trade, as it only increases the cost to reposition in the limit order book. If an order cancellation fee increases the cost to withdraw or revise limit order submissions, then traders may shift to marketable orders in an attempt to pick off the more static limit orders. The decrease in non-marketable orders and increase in marketable orders might increase the rate of transaction. We address this question by examining market share across the exchanges during the sample period. Specifically, we estimate market share on a daily basis for each option series as the number of trades on a particular exchange divided by the total number of trades across all exchanges. We test for differences across exchanges, before and after the fee change, using standard errors clustered by option series. We report the results of this market share analysis in Table 5.

[Insert Table 5 Here]

In Panel A of Table 5, we find that the average market share on the PHLX increases by 0.45 percentage points from the pre-fee period to the post-fee period. This increase in market share on the PHLX is statistically significant at the 0.01 level. However, we find even larger increases in market share on the AMEX (0.78 percentage points) and the ISE (1.85 percentage points), both of which are highly significant. It appears that the PHLX, AMEX, and ISE are capturing market share away from the CBOE and NYSE ARCA. While these results provide some evidence that the order cancellation fee on the PHLX creates a more attractive trading environment, it is difficult to make meaningful inferences due to the large changes in market share on competing venues.

Option market participants must comply with the trade through rule, which requires marketable orders to execute at or within the NBBO. As a result, the set of venues on which marketable orders will execute upon arrival is restricted to those offering to sell (buy) options at the NBO (NBB). Therefore, in Panel B of Table 5, we focus on trades which execute at the NBB (NBO) when each of the seven options exchanges, excluding BATS, has a bid (offer) price that is equal to the NBB (NBO).¹³ We find that the PHLX market share increases by 2.79 percentage points from the pre-fee period to the post-fee period. This increase is both statistically significant and economically meaningful. Our findings indicate that the PHLX captures market share away from the other PFOF venues (i.e. AMEX, CBOE, and NYSE ARCA), which is important because

¹³ Our decision to exclude BATS is due to its lack of quotes at the NBBO.

the relative take fee on a venue can affect marketable order routing decisions (see Battalio, Corwin, and Jennings, 2016).

We continue to find a large increase in market share on the ISE (2.63 percentage points) over the sample period. This might be the result of the ISE eliminating order restrictions in September of 2010, which permit traders other than market makers to enter orders on both sides of the market more freely. In the basis section of the proposed rule filing (see SEC Release No. 34-63017), the ISE argues that the changes will increase order flow and improve liquidity. This is, of course, just one possible explanation for the increase in trade volume on the ISE over the sample period. Nevertheless, the increase in market share on the PHLX, appears to be higher after the fee change, or at least not lower. Therefore, despite the increase in transactions costs, traders find it more, or just as appealing, to submit marketable orders to the PHLX after the cancellation fee is introduced. Since the limit orders on the PHLX book after the fee change remain visible for a longer period of time, fast liquidity demanders might be willing to absorb higher transactions costs in an attempt to pick-off the stale limit orders (Menkveld and Zoican, 2017). We formally examine this conjecture in the proceeding analysis.

5. Potential Explanation: Limit Order Pick-Off Risk

In this section, we examine if pick-off risk is an economic channel through which the limit order cancellation fee affects order flow and execution quality. Traders can manage pick-off risk by quickly canceling and revising their limit orders as new information (private or public) enters the marketplace. However, if an order cancellation fee increases monitoring costs, it can be more expensive to withdraw and revise limit order submissions. Consequently, traders may shift to marketable orders in an attempt to pick off the stale non-marketable limit orders. To test the validity of this argument, we examine whether pick-off risk increases on the PHLX, relative to the NOM, after the order cancellation fee is introduced.

Berkman (1996) shows that limit orders for options are picked off after adverse changes in the underlying stock price by examining option midquotes after a trade. Therefore, to proxy pick-off risk, we estimate the 5- to 15-minute price impact of order fills on the PHLX and NOM. Specifically, we estimate percent price impact (*% Price Impact*) as: $2D_k(M_{k+5} - M_k)/M_k$, where D_k equals +1 for a buy order and -1 for a sell order, M_{k+5} is the NBBO midquote 5- to 15-minutes following the order fill, and M_k is the NBBO midpoint at the time of the order fill. An advantage

of the limit order data is that we have an indicator for trade direction. Since Hasbrouck and Saar (2001) state that higher volatility increases the pick-off risk of limit orders, we also proxy pick-off risk using two volatility measures. The first is the standard deviation of intraday midpoint trade quotes (*Midpoint Volatility*) and the second is the log difference between the daily high ask price and the daily low bid price (*Range Volatility*).

[Insert Figure 4 Here]

Figure 4 plots average daily price impact and range volatility for option series on the PHLX (dark solid line) and the NOM (light dotted line) during the sample period. To smooth out the plots on pick-off risk, we estimate 10-day moving averages. Panel A of Figure 4 shows that limit order percent price impact increases for options on the PHLX and decreases for option on the NOM around the fee change. Panel B of Figure 4 shows that average daily range volatility increases for options on the PHLX and decreases for options on the NOM after the order cancellation fee is introduced, albeit the changes around the event are less discrete. These illustrations seem to suggest that the level of pick off risk is higher on the PHLX, relative to the NOM, after the fee change.

To statistically test whether pick-off risk increases on the PHLX after the introduction of the order cancellation fee, we estimate specifications of the following regression equation on option series/day observations:

$$\begin{aligned}
 \text{Pick} - \text{Off Risk}_{i,j,t} &= \alpha + \beta_1 \text{Phlx}_j + \beta_2 \text{Post}_t + \beta_3 \text{Phlx} \times \text{Post}_{j,t} \\
 &+ \beta_4 \text{Days} - \text{to} - \text{Expiration}_{i,j,t} + \beta_5 S/X_{i,j,t} + \beta_6 \text{Trade Price}_{i,j,t} \\
 &+ \beta_7 \text{Fragmentation}_{i,j,t} + \beta_8 \text{Underlying MCAP}_{i,t} \\
 &+ \beta_9 \text{Log}(\text{Underlying Volume})_{i,t} + \delta_t + \gamma_i + \varepsilon_{i,j,t},
 \end{aligned} \tag{3}$$

where the dependent variable is one of three pick-off risk measurements: % *Price Impact*, *Midpoint Volatility*, or *Range Volatility*. *Phlx* is a categorical variable equal to one if the order was initiated on the PHLX and zero if the order was initiated on the NOM. *Post* is a dummy variable equal to one if the order is submitted in the post-fee period and zero otherwise. The interaction term between *Phlx* and *Post* is the DID estimator. The remaining independent variables have previously been defined. In the full model specification, we include option fixed effects and day fixed effects. We report the results of estimating equation (3) in Table 6, with *t*-statistics in parentheses calculated from standard errors two-way clustered by option series and day.

[Insert Table 6 Here]

The results in the first two columns of Table 6 show that the average price impact increases for limit orders on the PHLX, relative to those on the NOM, after the introduction of the cancellation fee. In economic terms, the average limit order price impact on the PHLX increases between 6.78 and 6.93 percentage points, relative to the NOM, after the cancellation fee is introduced. In the third and fourth columns of Table 6, we find that, relative to an option on the NOM, the average midpoint volatility for an option on the PHLX increases between 0.1174 and 0.1122 after the fee is introduced. These results represent either a 56.55% or 54.05% increase above pre-fee volatility levels on the PHLX. Similar results are found in the final two columns of Table 6, where range-based volatility is the dependent variable. For instance, the average range volatility for an option on the PHLX, relative to an option on the NOM, increases between 0.0929 and 0.0952 after the order cancellation fee is introduced.

The results in this subsection suggest that the level of pick-off risk increases on the PHLX, relative to the NOM, after the cancellation fee is introduced. Since a fee on order cancellations can make it more difficult for traders to manage the risk of getting undercut while sitting on the limit order book, displayed limit orders might be more exposed to pick-off risk. Consequently, traders might prefer marketable orders to non-marketable order in the post-fee trading environment, which can affect various aspects of execution quality.

6. Robustness

In this section, we report the results of a series of robustness tests to help validate our findings. First, we perform a pseudo-event study, where we examine order cancellation activity for options on the PHLX around an alternative event date. We select the calendar year immediately following the event date, August 19, 2011. We find that the coefficient on the categorical variable *Post*, in the cancellation regression is insignificant, providing support for our main analysis. Next, we examine if the cancellation fee results in a reduction in low-latency order cancellation activity (see Section A in the Internet Appendix). Following Hasbrouck and Saar (2013), we identify linked cancellation messages over 10-minute intervals throughout the day that are likely to be a part of the same algorithmic strategy. Our results confirm our previous findings that cancellation activity decreases after the fee is introduced, i.e. fewer runs and shorter run times. Third, we examine if there are differential effects across option intrinsic values and contract types (call vs. put) around

the implementation of the order cancellation fee on the PHLX.¹⁴ We estimate a series of triple difference regressions, but do not find meaningful differences between option moneyness categories or contract types. However, we do find that in-the-money put option further from expiration are more likely to have high order cancellation rates ex-post (see Section C in the Internet Appendix).

We also examine if our main results hold in a subsample of ETF option series (see Section D in the Internet Appendix). The results are generally the same between options on common stocks and options on ETFs. Additionally, to obtain bias-corrected intervals around the coefficients in our main regression equations (1) – (3), we conduct a Bootstrap analysis (see Section E in the Internet Appendix). We use 1,000 random sampling replications to assign measures of accuracy to the sample estimates. We find that the regression coefficients are nearly identical to those reported in the previous sections. In our final robustness check, we conduct a linear Bayesian analysis to more confidently assign causality (see Section E in the Internet Appendix). The assumption is that all model parameters are random quantities. We use Gibbs sampling and 10,000 Markov chain Monte Carlo (MCMC) simulations on simplified forms of our main regression equations (1) - (3). The results are similar to those reported in the previous sections.

7. Concluding Remarks

Limit orders play a pivotal role in options markets (see Berkman, 1996). The canceling of these limit orders has captured significant attention from exchange officials, the popular press, and regulators. In this study, we examine order flow and execution quality around the introduction of an order cancellation fee on the PHLX. We use the absence of a similar cancellation fee on the NOM as a control group. We find that the cancellation fee reduces the number of orders that are submitted and subsequently deleted. Our results also show that the order cancellation fee increases the average duration of resting limit orders on the book, which might reduce the concern with “phantom liquidity”, or quotes that disappear before they are traded against (see Angel, 2014; and Friederich and Payne, 2015). However, the inability for liquidity supplying traders to quickly cancel and revise their order submissions without incurring substantial fees, might expose their order flow to being picked off by future traders with access to new information.

¹⁴ Empirical research highlights differences in trading costs, open interest, and trading volume across option features (see Lakonishok et al., 2007; Roll, Schwartz, and Subrahmanyam, 2010; Battalio, Shkilko, and Van Ness, 2016).

We find that the number of non-marketable orders submitted to the PHLX significantly declines after the fee is introduced. Additionally, the percentage of professional non-marketable orders declines by 17.74 percentage points from the pre-fee period to the post-fee period. We contend that many liquidity-providing professional traders leave the PHLX market altogether after the fee is implemented. The decline in liquidity provision, and increase in pick-off risk, leads to higher trading costs for liquidity demanders. Specifically, dollar effective spreads increase by approximately 14.93% after the fee is introduced, other factors held constant. The increase in trading costs, however, does not deter market participation, as the arrival of marketable orders increases, and the PHLX captures additional market share, or at least does not lose market share. These results seem to suggest that liquidity demanders are willing to pay slightly higher transactions costs in an attempt to pick off the stale non-marketable limit orders.

Our analysis also reveals that the average non-marketable order fill rate increases by 7.3 percentage points. To the extent that limit order traders are better off when facing less non-execution risk (Colliard and Foucault, 2012), lower cancellation activity seems to have a positive impact on trader welfare. Furthermore, the average time-to-complete fill on marketable limit orders is 9.5 milliseconds faster in the post-fee period. Speed is an important aspect of execution quality, as Blume (2002) cites a survey conducted by an investment firm, which found that 58 percent of online traders rated immediacy of execution higher than a favorable price. Exchange latency enables fast liquidity demanders to profit by trading against potentially stale quotes (Menkveld and Zoican, 2017).

The implications of our results are broad, as exchange officials in both equity and options markets are addressing the potential problems associated with limit order cancellations. Competing trading venues with similar market structures to that of the PHLX, might consider adopting a targeted order cancellation fee.

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Table 1. Sample Selection

The table summarizes the sample selection process for active equity option series on the PHLX between July 26, 2010 and October 18, 2010. An option series refers to a unique underlying asset, call or put, expiration date, and exercise (strike) price. The final sample consists of 7,492 option series on 116 underlying securities.

<i>Panel A. Sample filters</i>	
	<i># of remaining option series</i>
Initial sample	143,670
Exclude option series with fewer than 1 order fill per day	33,136
Exclude option series with fewer than 5 orders in a day	15,569
Merge option series on PHLX and NOM by option series day	8,844
Exclude option series that are not on common stocks or on ETFs	7,492
Final Sample	7,492
<i>Panel B. Distribution of option series</i>	
	<i># of option series</i>
Option series on common stocks	5,850
Option series on ETFs	1,642
Final Sample	7,492

Table 2. Order Cancellation Fees and Order Flow

The table reports the results from estimating specifications of the following fixed effects regression equation on average daily option series observations:

$$Order\ Flow_{i,j,t} = \alpha + \beta_1 Phlx_j + \beta_2 Post_t + \beta_3 Phlx \times Post_{j,t} + \beta_4 Days - to - Expiration_{i,j,t} + \beta_5 S/X_{i,j,t} + \beta_6 Order\ Price_{i,j,t} + \beta_7 Fragmentation_{i,j,t} + \beta_8 Underlying\ MCAP_{i,t} + \beta_9 Log(Underlying\ Volume)_{i,t} + \delta_t + \gamma_i + \varepsilon_{i,j,t}.$$

The dependent variable is set to one of the following order flow measures: *Order Cancellation Rate*, *Order Duration*, *Log (# of Non-Marketable Orders)*, or *Log (# of Marketable Orders)*. *Order Cancellation Rate* equals the number of orders canceled divided by total orders submitted. *Order Duration* equals the number of seconds between order submission and cancellation. We also examine the natural log of the total number of non-marketable and marketable orders submitted to the PHLX and NOM. *Phlx* is an indicator variable set equal to one if the order originates on the PHLX and zero for orders originating on the NOM. *Post* equals one if the order is submitted after the introduction of the PHLX order cancellation fee and zero otherwise. *Days-to-Expiration* equals the number of trading days between the order date and option expiration date. *S/X* equals the underlying stock price divided by the option strike price. *Order Price* is the average daily order price. *Fragmentation* is measured as volume executed outside of the trade exchange (in thousands). *Underlying MCAP* is the market capitalization (in \$billions) of the underlying security. *Underlying Volume* equals the total number of shares traded in the underlying security. We also include both day fixed effects and option series fixed effects. We report *t*-statistics in parentheses obtained from standard errors two-way clustered by option series and day. ***, **, denote statistical significance at the 0.01 and 0.05 levels, respectively.

	Order Cancellation Rate		Order Duration (seconds)		Log (# of Non-Marketable Orders)		Log (# of Marketable Orders)	
	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
Phlx	-0.2989*** (-30.7467)	-0.2988*** (-30.8724)	653.4851*** (23.0068)	656.0457*** (23.3680)	-4.8960*** (-84.3678)	-4.8948*** (-84.4500)	-0.1138*** (-2.9185)	-0.1643*** (-4.3148)
Post	0.0005 (0.8089)		12.2689 (1.6654)		-0.1544** (-2.4972)		0.0576 (1.0617)	
Phlx × Post	-0.1197*** (-9.8353)	-0.1197*** (-9.9004)	217.4798*** (6.1146)	218.5647*** (6.1950)	-0.8246*** (-9.8085)	-0.8252*** (-9.8644)	0.1309*** (2.9592)	0.1582*** (3.5499)
Days-to-Expiration		0.0002 (1.1483)		-4.8739*** (-6.9034)		-0.0090*** (-7.5431)		-0.0200*** (-11.5426)
S/X		0.0106 (0.3584)		-257.6071 (-1.3462)		0.0878 (0.4422)		-0.5248 (-1.9011)
Order Price		0.0018*** (4.6609)		-1.0835 (-0.8044)		0.0201*** (5.8046)		-0.0368*** (-7.5729)
Fragmentation		0.0001*** (2.7188)		-1.2696** (-2.1747)		0.0026** (2.0098)		0.0050 (1.6734)
Underlying MCAP		-0.0009 (-1.8977)		1.8528 (1.1457)		0.0014 (0.4115)		0.0152*** (3.8809)
Log(Underlying Volume)		-0.0074 (-1.9240)		-75.3995*** (-4.0391)		0.2938*** (10.1991)		0.4931*** (14.1448)
Constant	0.9956*** (149.2026)		79.8489*** (12.0631)		8.9495*** (173.2784)		2.6721*** (55.2905)	
Day FE	No	Yes	No	Yes	No	Yes	No	Yes
Option FE	No	Yes	No	Yes	No	Yes	No	Yes
R ²	0.4793	0.5544	0.1077	0.1288	0.7348	0.8550	0.0017	0.0654
N	65,760	65,760	64,971	64,904	65,760	65,760	36,226	35,076

Table 3. Order Cancellation Fees and Order Flow: by Trader Type

The table reports order flow on the PHLX separated by trader type (professional, broker-dealer, customer, firm, and market maker). We report daily observations averaged across option series. Test-statistics are used to determine if order flow differs across the introduction of the order cancellation fee. Standard errors are two-way clustered by option series and day. ***, ** denote statistical significance at the 0.01 and 0.05 levels, respectively.

	% of All Orders	% of Non-Marketable Orders	% of Marketable Orders	Non-Marketable Fill Rate	Non-Marketable Cancellation Rate
<i>Panel A. Professional Order Flow</i>					
Pre	42.38%	42.90%	4.13%	0.77%	98.44%
Post	24.44%	25.16%	4.13%	1.77%	96.60%
Difference (Post - Pre)	-17.94%***	-17.74%***	0.00%	1.00%***	-1.84%***
t-stat	(-10.852)	(-10.742)	(-0.009)	(5.542)	(-6.721)
<i>Panel B. Broker-Dealer Order Flow</i>					
Pre	1.71%	1.68%	1.63%	14.44%	73.88%
Post	1.10%	1.08%	1.72%	22.42%	64.15%
Difference (Post - Pre)	-0.61%*	-0.60%*	0.09%	7.98%***	-9.73%***
t-stat	(-1.954)	(-1.941)	(0.495)	(4.052)	(-3.634)
<i>Panel C. Customer Order Flow</i>					
Pre	49.59%	49.36%	85.24%	38.92%	36.98%
Post	68.62%	68.28%	84.75%	37.54%	39.07%
Difference (Post - Pre)	19.03%***	18.92%***	-0.49%	-1.38%***	2.09%***
t-stat	(12.193)	(12.093)	(-0.542)	(-2.740)	(5.344)
<i>Panel D. Firm Order Flow</i>					
Pre	6.11%	5.88%	8.28%	21.48%	62.29%
Post	5.55%	5.25%	8.47%	27.64%	55.19%
Difference (Post - Pre)	-0.56%	-0.63%	0.19%	6.16%***	-7.10%***
t-stat	(-0.884)	(-1.003)	(0.306)	(3.279)	(-3.141)
<i>Panel E. Market Maker Order Flow</i>					
Pre	0.20%	0.18%	0.71%	9.92%	65.03%
Post	0.27%	0.24%	0.91%	13.92%	62.27%
Difference (Post - Pre)	0.07%*	0.06%	0.20%	4.00%	-2.76%
t-stat	(1.678)	(1.337)	(1.424)	(1.436)	(-0.620)

Table 4. Order Cancellation Fees and Order Execution Quality

The table reports the results from estimating specifications of the following fixed effects regression equation on average daily option series observations:

$$\text{Order Execution Quality}_{i,j,t} = \alpha + \beta_1 \text{Phlx}_j + \beta_2 \text{Post}_t + \beta_3 \text{Phlx} \times \text{Post}_{j,t} + \beta_4 \text{Days} - \text{to} - \text{Expiration}_{i,j,t} + \beta_5 S/X_{i,j,t} + \beta_6 \text{Price}_{i,j,t} + \beta_7 \text{Fragmentation}_{i,j,t} + \beta_8 \text{Underlying MCAP}_{i,t} + \beta_9 \text{Log}(\text{Underlying Volume})_{i,t} + \delta_t + \gamma_i + \varepsilon_{i,j,t}.$$

The dependent variable is set to one of the following order execution quality measures: *Order Fill Rate*, *Order Fill Speed*, or *\$ Effective Spread*. *Order Fill Rate* equals the number of non-marketable orders completely filled to total orders submitted. *Order Fill Speed* is the number of seconds between order submission and complete fill, estimated separately for marketable and non-marketable orders. *\$ Effective Spread* equals twice the absolute difference between the trade price and the BBO midpoint on a particular exchange. *Phlx* is an indicator variable set equal to one if the order originates on the PHLX and zero otherwise. *Post* equals one if the order is submitted after the introduction of the PHLX order cancellation fee and zero otherwise. *Days-to-Expiration* equals the number of trading days between the order date and option expiration date. *S/X* equals the underlying stock price divided by the strike price. *Price* is either the average daily order price (order fill rate/speed) or trade price (effective spread). *Fragmentation* is measured as volume executed outside of the trade exchange (in thousands). *Underlying MCAP* is the market capitalization (in \$billions) of the underlying security. *Underlying Volume* equals the total number of shares traded in the underlying security. We also include both day fixed effects and option series fixed effects. We report *t*-statistics in parentheses obtained from standard errors two-way clustered by option series and day. ***, **, denote statistical significance at the 0.01 and 0.05 levels, respectively.

	Order Fill Rate (PHLX vs. NOM)		Order Fill Speed (PHLX vs. NOM)				\$ Effective Spread (PHLX vs. Market)	
			Marketable		Non-marketable			
	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
Phlx	0.1684*** (28.3709)	0.1684*** (28.4906)	-0.0183*** (-8.4570)	-0.0175*** (-7.4669)	318.7834*** (9.3530)	308.3996*** (9.2891)	-0.0175*** (-10.2296)	-0.0185*** (-10.6197)
Post	-0.0007 (-1.2719)		0.0055** (2.3136)		17.2918 (0.3453)		-0.0016 (-0.4184)	
Phlx × Post	0.0730*** (9.4836)	0.0730*** (9.5436)	-0.0101*** (-3.9847)	-0.0095*** (-3.5339)	3.0213 (0.0781)	4.7434 (0.1274)	0.0072*** (3.6450)	0.0079*** (3.9474)
Days-to-Expiration		-0.0000 (-0.0935)		-0.0004*** (-5.6017)		-5.6668*** (-4.2512)		0.0000 (0.2095)
S/X		0.0277 (1.6982)		-0.0066 (-0.6604)		-1006.3926*** (-3.7476)		-0.0829*** (-4.2829)
Price		-0.0011*** (-3.6842)		-0.0006*** (-3.1620)		-0.9063 (-0.2813)		0.0053*** (11.6659)
Fragmentation		0.0266 (0.6762)		0.0000 (1.3204)		-3.5478** (-2.1325)		0.0001 (1.7914)
Underlying MCAP		0.0005 (1.5799)		0.0001 (0.9963)		1.7747 (0.7049)		-0.0012*** (-6.0612)
Log(Underlying Volume)		0.0172*** (6.0964)		0.0048*** (3.4818)		-191.4939*** (-5.6573)		0.0111*** (6.2215)
Constant	0.0035*** (6.3222)		0.0522*** (26.9668)		943.6244*** (24.0834)		0.0704*** (22.7675)	
Day FE	No	Yes	No	Yes	No	Yes	No	Yes
Option FE	No	Yes	No	Yes	No	Yes	No	Yes
R ²	0.3713	0.4280	0.0257	0.0360	0.0076	0.0205	0.0011	0.0303
N	65,760	65,760	36,226	35,076	61,851	61,350	186,335	186,335

Table 5. Market Share of Trades across Exchanges

Panel A. The table reports the market share of trades in common stock option series across all eight exchanges around the introduction of the PHLX cancellation fee. Test-statistics are reported in parentheses obtained from standard errors two-way clustered by option series and day. ***, ** denote statistical significance at the 0.01 and 0.05 levels, respectively.

	NOM	AMEX	CBOE	ISE	NYSE ARCA	PHLX	BOX	BATS
Pre	13.17%	11.60%	23.44%	16.77%	12.95%	14.73%	6.07%	3.38%
Post	13.46%	12.38%	20.90%	18.62%	11.59%	15.17%	6.28%	3.26%
Difference (Post - Pre)	0.29%	0.78%***	-2.54%***	1.85%***	-1.36%***	0.45%***	0.21%	-0.12%
<i>t</i> -stat	(1.25)	(4.38)	-(12.05)	(8.61)	-(8.05)	(2.60)	(1.91)	-(1.34)

Panel B. The table reports the market share of trades in common stock option series when each of the seven exchanges are at the best posted quote (NBBO) around the introduction of the PHLX cancellation fee. We exclude BATS from the analysis due to its lack of quotes at the NBBO. Test-statistics are reported in parentheses obtained from standard errors two-way clustered by option series and day. ***, ** denote statistical significance at the 0.01 and 0.05 levels, respectively.

	NOM	AMEX	CBOE	ISE	NYSE ARCA	PHLX	BOX
Pre	12.61%	16.54%	20.49%	17.91%	12.17%	10.39%	9.89%
Post	12.66%	15.14%	18.97%	20.54%	9.97%	13.18%	9.54%
Difference (Post - Pre)	0.05%	-1.40%***	-1.52%***	2.63%***	-2.20%***	2.79%***	-0.35%
<i>t</i> -stat	(0.10)	-(2.60)	-(2.53)	(3.11)	-(4.31)	(5.08)	-(0.67)

Table 6. Order Cancellation Fees and Pick-Off Risk

The table reports the results of estimating specifications of the following regression equation on average daily option series observations:

$$\begin{aligned} \text{Pick - Off Risk}_{i,j,t} = & \alpha + \beta_1 \text{Phlx}_j + \beta_2 \text{Post}_t + \beta_3 \text{Phlx} \times \text{Post}_{j,t} + \beta_4 \text{Days - to - Expiration}_{i,j,t} + \\ & \beta_5 S/X_{i,j,t} + \beta_6 \text{Trade Price}_{i,j,t} + \beta_7 \text{Fragmentation}_{i,j,t} + \beta_8 \text{Underlying MCAP}_{i,t} + \\ & \beta_9 \text{Log(Underlying Volume)}_{i,t} + \delta_t + \gamma_i + \varepsilon_{i,j,t}. \end{aligned}$$

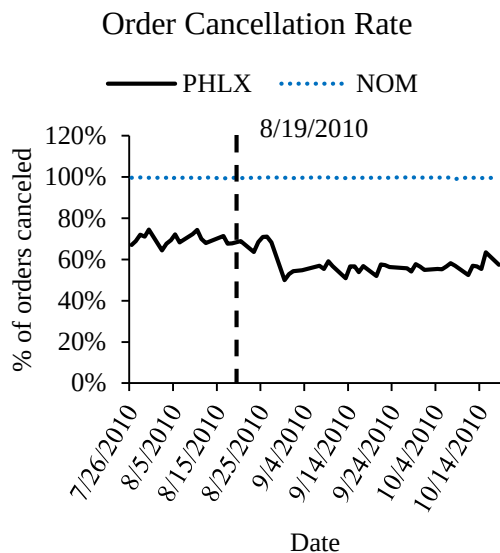
The dependent variable is set to one of three measures of pick-off risk: % Price Impact, Midpoint Volatility, or Range Volatility. % Price Impact is measured by the following equation: $2D_k(M_{k+5} - M_k)/M_k$, where D_k equals +1 for a buy order and -1 for a sell order, M_{k+5} is the NBBO midpoint 5- to 15-minutes following the limit order fill, and M_k is the NBBO midpoint at the time of the limit order fill. Midpoint Volatility is equal to the standard deviation of intraday midpoint quotes. Range Volatility is equal to the log difference between the daily high ask price and the daily low bid price. Phlx is an indicator variable set equal to one if the order originates on the PHLX and zero for orders originating on the NOM. Post equals one if the trade executes after the introduction of the PHLX order cancellation fee and zero otherwise. Days-to-Expiration equals the number of trading days between the order date and option expiration date. S/X equals the underlying stock midpoint price at the time of the trade divided by the strike price. Trade Price is the average daily trade price. Underlying MCAP is the underlying stock's average daily market capitalization, measured in \$billions. Fragmentation is measured as volume executed outside of the trade exchange (in thousands). Underlying Volume equals the underlying stock's average daily share volume. We also include both day fixed effects and option series fixed effects. We report *t*-statistics in parentheses obtained from standard errors two-way clustered by option series and day. ***, **, denote statistical significance at the 0.01 and 0.05 levels, respectively.

	% Price Impact		Midpoint Volatility		Range Volatility	
	[1]	[2]	[3]	[4]	[5]	[6]
Phlx	-0.0238 (-1.9379)	-0.0203 (-1.6830)	-0.1122*** (-9.1937)	-0.1125*** (-9.3655)	-0.0886*** (-8.4651)	-0.0890*** (-7.9130)
Post	-0.0378*** (-3.4710)		-0.0940*** (-4.2151)		-0.0643*** (-3.6153)	
Phlx × Post	0.0678*** (4.1785)	0.0693*** (4.1935)	0.1174*** (9.3272)	0.1122*** (9.1342)	0.0929*** (8.4446)	0.0952*** (8.2022)
Days-to-Expiration		-0.0001 (-0.1911)		0.0017*** (3.4573)		-0.0040*** (-7.2671)
S/X		-0.1859 (-0.7608)		0.3524*** (4.4059)		-0.0038 (-0.0394)
Trade Price		-0.0013 (-0.6179)		0.0197*** (10.2465)		-0.0120*** (-9.3452)
Fragmentation		0.0000 (0.1072)		0.0001 (1.2812)		0.0011 (1.7284)
Underlying MCAP		0.0011 (0.5777)		-0.0032 (-1.7051)		0.0001 (0.0519)
Log(Underlying Volume)		-0.0089 (-0.4583)		0.1573*** (11.5707)		0.1550*** (10.1814)
Constant	0.0301*** (4.4244)		0.3198*** (16.3601)		0.3835*** (25.1402)	
Day FE	No	Yes	No	Yes	No	Yes
Option FE	No	Yes	No	Yes	No	Yes
R ²	0.0003	0.0013	0.0078	0.1331	0.0037	0.0759
N	55,063	54,099	55,063	54,099	55,063	54,099

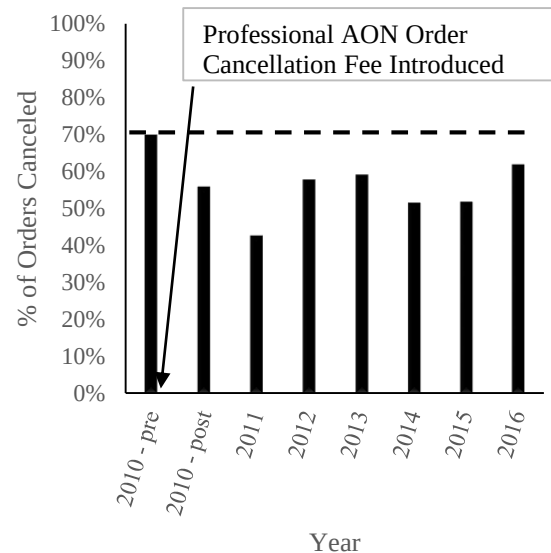
Figure 1. Order Cancellation Fees and Order Cancellation Activity

Panel A of Figure 1 plots average daily order cancellation rates for option series trading on the PHLX (dark solid line) and the NOM (light dotted line) around the event date, August 19th, 2010. Panel B of Figure 1 plots average daily order cancellation rates for active option series on the PHLX from the period 2010 to 2016. We examine orders submitted and canceled during the month of September in each year. Panel C of Figure 1 plots average daily order cancellation rates in September of 2016 across four exchanges: PHLX, Bats BZX, Bats EDGX, and NOM.

Panel A. Order Cancellation Rates – Event Study



Panel B. PHLX Order Cancellation Rates – Longitudinal Study



Panel C. 2016 Order Cancellation Rates by Exchange

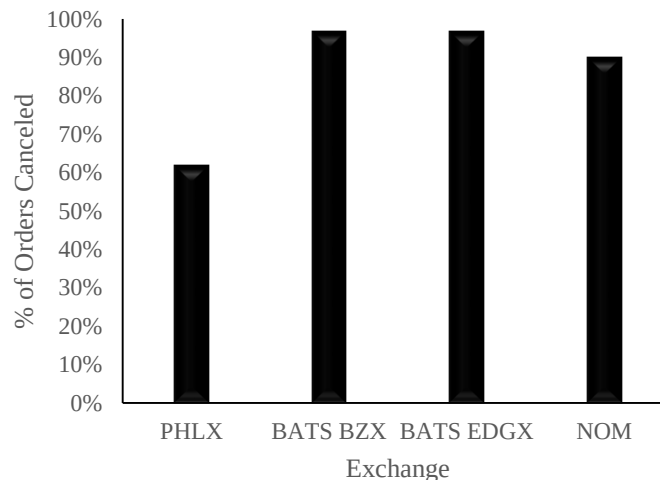


Figure 2. Order Cancellation Fees and Order Flow

Figure 2 plots average daily order duration and non-marketable/marketable order flow for option series trading on the PHLX (dark solid line) and the NOM (light dotted line) around the event date, August 19th, 2010. In Panel A, we plot average daily *Order Duration*, or the number of seconds between order submission and cancellation, around the fee change. In Panels B and C, we plot the average daily natural log of both the number of non-marketable and marketable order submissions around the fee change.

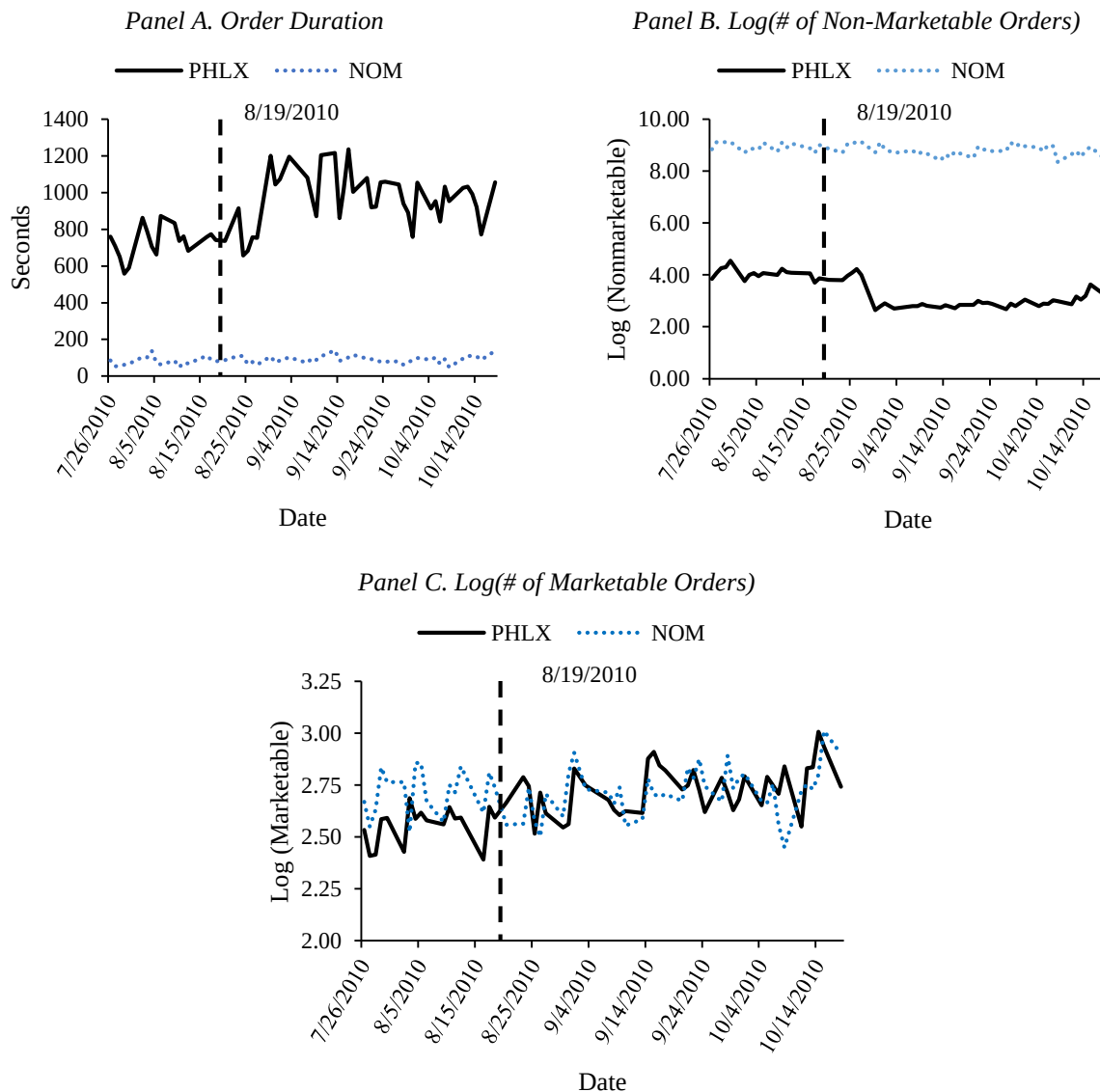


Figure 3. Order Cancellation Fees and Order Execution Quality

Panels A through C of Figure 3 plot average daily order fill rates and fill speeds for option orders on the PHLX (dark solid line) and the NOM (light dotted line) in event time. Panel D of Figure 3 plots average daily dollar effective spreads for option trades on the PHLX (dark solid line) and the other seven options exchanges (light dotted line) in event time. *Order Fill Rate* equals the number of non-marketable orders completely filled to total orders submitted. *Order Fill Speed* is the number of seconds between order submission and complete fill, estimated separately for marketable and non-marketable orders. *\$ Effective Spread* equals twice the absolute difference between the trade price and the BBO midpoint on a particular exchange. We plot simple averages for order fill rates and 10-day moving averages for order fill speeds and effective spreads.

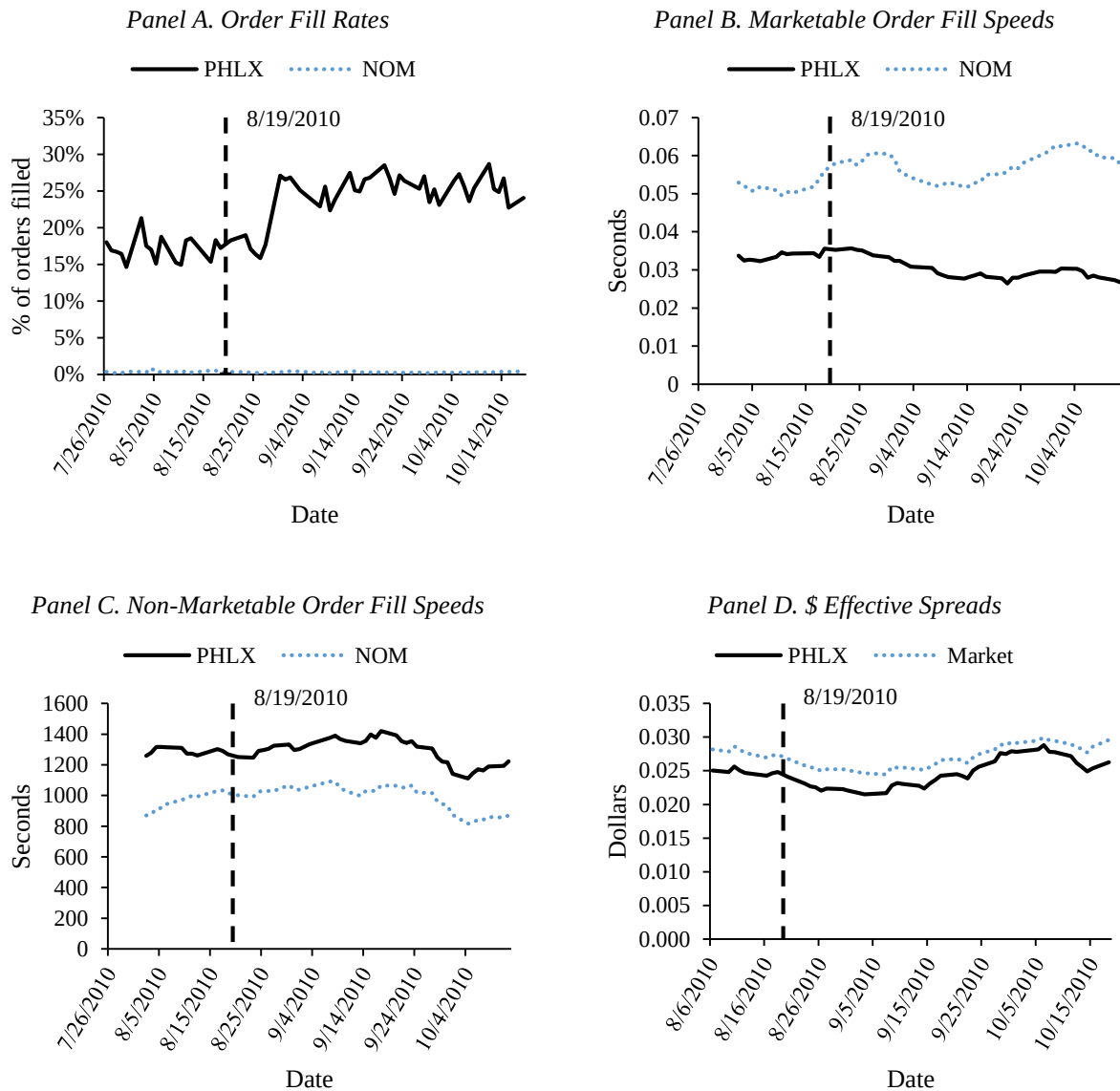
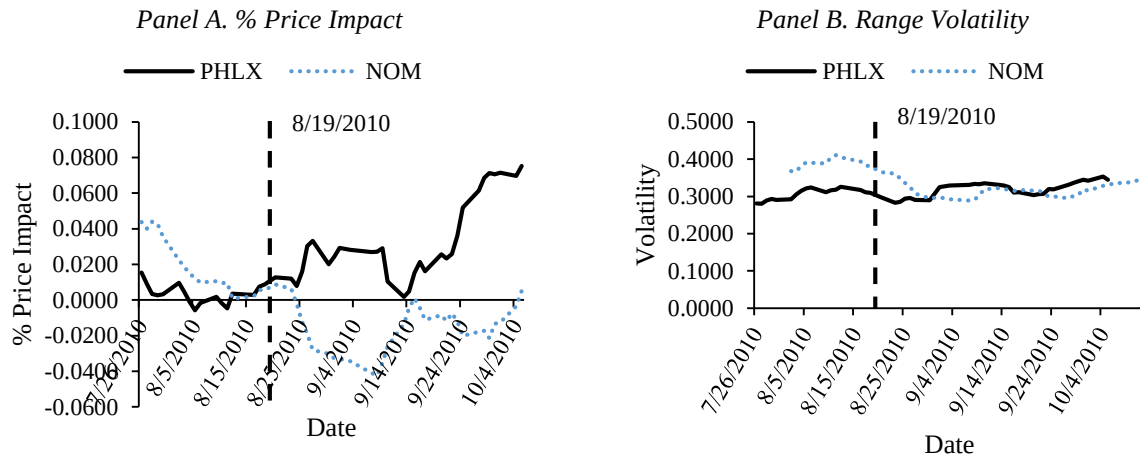


Figure 4. Order Cancellation Fees and Pick-Off Risk

Figure 4 plots average daily percent price impact and range volatility for option series trading on the PHLX (dark solid line) and the NOM (light dotted line) around the event date, August 19th, 2010. % *Price Impact* is measured by the following equation: $2D_k(M_{k+5} - M_k)/M_k$, where D_k equals +1 for a buy order and -1 for a sell order, M_{k+5} is the NBBO midpoint 5- to 15-minutes following the limit order fill, and M_k is the NBBO midpoint at the time of the limit order fill. *Range Volatility* is equal to the log difference between the daily high ask price and the daily low bid price. We plot 10-day moving averages for both figures.



Order Cancellations, Fees, and Execution Quality in U.S. Equity Options

Internet Appendix

A. *Low-Latency Order Cancellations*

As daily order cancellation rates are a crude measure of intraday cancellation activity, we also examine if the cancellation fee results in a reduction in low-latency order cancellation activity. Following Hasbrouck and Saar (2013), we identify linked cancellation messages over 10-minute intervals throughout the day that are likely to be a part of the same algorithmic strategy. We define a run as 5 or more messages for the same option series, at the same price, on the same side of the order book (buy or sell), and for the same order volume that are canceled during a given 10-minute interval. We then time-weight the number of runs during the interval and term it *RunsInCancellations*. We report the results of this analysis in Table A.1. Our results confirm our previous findings that cancellation activity decreases after the fee is introduced. Specifically, the number of *RunsInCancellations* decreases by -0.2096 from the pre-fee period to the post-fee period, which is significant at the 0.01 level. We also find a significant decrease in raw runs (not time-weighted) and a decrease in the average run time (in seconds).

B. *Order Cancellation Fees and Dollar Volume*

In section 4.3 of the manuscript, we provide evidence of an increase in trading activity on the PHLX around the fee change. However, as other option venues experience an increase in market share during the sample period, it is difficult to make meaningful inferences. For this reason, we also examine daily dollar volume on the PHLX, relative to the other seven options exchanges, surrounding the change in cancellation fees. Figure B.1 plots average daily dollar volume for option series on the PHLX (dark solid line) and for options series across the other seven options exchanges (light dotted line). To smooth out the plot, we estimate 10-day rolling averages. We show that the average daily dollar volume appears to increase more on the PHLX, relative to the other seven options exchanges, after the introduction of the cancellation fee, but at a gradual rate. The increase in trading volume across all options markets might be attributable to the unveiling of a monetary stimulus package at the end of August 2010 by former SEC Chairman Ben Bernanke.

To more accurately determine the effect of an order cancellation fee on dollar volume, controlling for cross-sectional and time-series variation, we estimate equation (2) in the manuscript but replace the dependent variable with dollar volume. We report the results of this analysis in Table B.1, with *t*-statistics in parentheses calculated from standard errors two-way clustered by option series and day. Our results show that average daily dollar volume is significantly higher on the PHLX, relative to the other seven options exchanges, after the cancellation fee is introduced. We note that the coefficient on the interaction term in the second column of Table B.1 is 58.9433, which is significant at the 0.05 level. In economic terms, the change in dollar volume represents an 8.14% increase on the PHLX above pre-event levels, relative to all other option venues, other factors held constant.

C. Nature of Option Contracts for which Cancellation Rates Remain High

In this section, we examine if there are differences in option contracts for which cancellation rates remain high in the post-fee trading environment. Since our analysis shows that an order cancellation fee affects order execution quality, it is important for market participants to know which contracts carry the highest risks. We estimate both a linear probability model and a logit on a dichotomous variable equal to one if the option series has an order cancellation rate above the 75th percentile (mean order cancellation rate = 97.72%), and zero if the order cancellation rate is below the 25th percentile (mean order cancellation rate = 26.11%).¹⁵ Since we are interested in the nature of the option contracts for which cancellation rates remain high, we restrict the analysis to observations on the PHLX and estimate the following DID model:

$$\begin{aligned}
 \text{High Cancellations}_{i,t} &= \alpha + \beta_1 \text{Call}_i \times \text{Post}_t + \beta_2 \text{Days-to-Expiration}_{i,t} \times \text{Post}_t \\
 &+ \beta_3 \text{In-the-Money}_{i,t} \times \text{Post}_t + \beta_4 \text{Penny}_i \times \text{Post}_t + \beta_5 \text{ETF}_i \times \text{Post}_t \\
 &+ \delta_t + \gamma_i + \varepsilon_{i,t},
 \end{aligned} \tag{C-1}$$

Post equals one if the order is submitted after the fee change and zero otherwise. *Call* equals one if the option is a call option and zero for a put option. *Days-to-Expiration* is the number of days until option expiration. *In-the-Money* equals one if the option is in-the-money and zero for out-of-the-money. *Penny* equals one if the option is quoted and traded in pennies and zero if the option is quoted and traded in nickels. *ETF* equals one if the option series is on an ETF and zero for a

¹⁵ We also identify high cancellations as observations in the 90th and 10th percentile levels and our results are similar.

common stock. We include both day fixed effects and option fixed effects. Each interaction term is a DID estimator between the pre-fee period and the post-fee period. We report the results of estimating equation (C-1) in Table C.1 with t -statistics in parentheses obtained from two-way clustered standard errors at the option and day levels.

We find that order cancellation rates are more likely to be elevated in the post-fee period for in-the-money options further from expiration. Specifically, the beta coefficient on the interaction term $In - the - Money_{i,t} \times Post_t$ is positive and significant, suggesting that in-the-money options are more likely to have high cancellation rates in the post-fee trading environment, relative to out-of-the-money options. The odds ratio suggests that in-the-money options are 1.8128 times more likely to have high order cancellation rates after the fee change, relative to out-of-the-money options. The positive and significant coefficient on $Days - to - Expiration_{i,t} \times Post_t$, indicates that options distant from expiration are 1.003 times more likely to have high order cancellation rates after the fee change, relative to near-term options. According to the fixed-effects logistic regression, put options are more likely to have high order cancellations ex-post, relative to call options, as the coefficients on the interaction term $Call_i \times Post_t$ is negative and significant at the 0.05 level. We also find that order cancellation rates are more likely to remain high for options on ETFs and non-penny options following the fee change.

D. Exchange Traded Funds (ETFs) Subsample

We analyze the generalizability of our results across assets by estimating equations (1) – (3) in the paper on a subsample of ETF options series. The results of this analysis are reported in Table D.1. The findings are generally the same between options on common stocks and options on ETFs, with a few exceptions. We do not find significant changes in the average marketable order fill speed on the PHLX, relative to the NOM, in ETF options after the introduction of the cancellation fee. We also do not find a significant change in % price impact for ETF options on the PHLX, relative to the NOM, after the introduction of the order cancellation fee.

E. Bootstrap and Bayesian Analyses

We also conduct a Bootstrap analysis, to obtain bias-corrected intervals around the coefficients in our main regressions (1) – (4). We use 1,000 random sampling replications to assign measures of accuracy to the sample estimates. The result of this analysis are reported in Table E.1.

For brevity, we report the estimates for the interaction term between *Phlx* and *Post* only. We find that the regression coefficients are nearly identical to those reported in sections 4 and 5 of the manuscript.

As trading activity and spreads are increasing for both the PHLX and the overall market, we conduct a linear Bayesian analysis to more confidently assign causality. The assumption is that all model parameters are random quantities. We use the posterior distribution, assigned by the Bayes rule, to form probabilistic summaries of the model parameters (i.e. means, medians, standard deviations, and credible intervals). We use Gibbs sampling and 10,000 Markov chain Monte Carlo (MCMC) simulations on simplified forms of our main regression equations (1) - (4). The results of this analysis are reported in Table E.2. We find that our results are robust. For instance, the mean estimates and standard deviations on the interaction term between *Phlx* and *Post* are similar to the means and standard errors reported in previous tables. However, we can now state that the probability that the average dollar effective spread on the PHLX is between 0.003 and 0.011, relative to the market, after the fee change is 0.95. Since the lower bound is greater than zero, we conclude that the cancellation fee has a positive and significant impact on dollar effective spreads. The remaining results in Table E.2 are similar to the regression results reported in sections 4 and 5 in the manuscript.

Table A.1. Order Cancellation Fees and Runs in Cancellations

The table reports runs in cancellation messages during ten-minute intervals around the introduction of the PHLX cancellation fee. *** denotes statistical significance at the 0.01 level.

	Raw Runs	RunsInCancellations	Run Time
Pre	4.3236	1.8995	267.4235
Post	4.0528	1.6899	246.1057
Difference (Post - Pre)	-0.2708***	-0.2096***	-21.3178***
<i>t</i> -stat	(-3.9273)	(-7.6007)	(-20.5427)

Table B.1. Order Cancellation Fees and Dollar Volume

The table reports the results from estimating specifications of the following regression equation on average daily option series observations:

$$\begin{aligned} \$Volume_{i,j,t} = & \alpha + \beta_1 Phlx_j + \beta_2 Post_t + \beta_3 Phlx \times Post_{j,t} + \beta_4 Days - to - Expiration_{i,j,t} + \beta_5 S/X_{i,j,t} \\ & + \beta_6 Trade\ Price_{i,j,t} + \beta_7 Fragmentation_{i,j,t} + \beta_8 Underlying\ MCAP_{i,t} + \\ & \beta_9 Log(Underlying\ Volume)_{i,t} + \delta_t + \gamma_i + \varepsilon_{i,j,t}. \end{aligned}$$

The dependent variable is *Dollar Volume*, or the number of contracts in a trade times the trade price. *Phlx* is an indicator variable set equal to one if the trade executes on the PHLX and zero for trades executed on one of the seven remaining option exchanges. *Post* equals one if the trade executes after the introduction of the PHLX order cancellation fee and zero otherwise. *Days-to-Expiration* equals the number of trading days between the trade date and option expiration date. *S/X* equals the underlying stock midpoint price at the time of the trade divided by the strike price. *Trade Price* is the average daily trade price. *Underlying MCAP* is the underlying stock's average daily market capitalization, measured in \$billions. *Fragmentation* is measured as volume executed outside of the trade exchange (in thousands). *Underlying Volume* equals the underlying stock's average daily share volume. We also include both day fixed effects and option series fixed effects. We report *t*-statistics in parentheses obtained from standard errors two-way clustered by option series and day. ***, **, denote statistical significance at the 0.01 and 0.05 levels, respectively.

	[1]	[2]
Phlx	161.3642*** (7.6315)	172.4830*** (7.5793)
Post	199.0981*** (3.1689)	
Phlx × Post	55.8036** (2.1441)	58.9433** (2.1416)
Days-to-Expiration		-14.1878*** (-9.1726)
S/X		-653.0276*** (-2.6328)
Trade Price		33.6801*** (3.6989)
Fragmentation		4.7484** (2.0777)
Underlying MCAP		37.3376*** (3.6064)
Log(Underlying Volume)		582.3347*** (12.2048)
Constant	563.1451*** (14.0314)	
Day FE	No	Yes
Option FE	No	Yes
R ²	0.0050	0.0868
N	186,335	186,335

Table C.1. Nature of Option Contracts for which Order Cancellation Rates Remain High

The table reports the results of estimating the following equation on average daily option series observations:

$$\text{High Cancellations}_{i,t} = \alpha + \beta_1 \text{Call}_i \times \text{Post}_t + \beta_2 \text{Days-to-Expiration}_{i,t} \times \text{Post}_t + \beta_3 \text{In-the-Money}_{i,t} \times \text{Post}_t + \beta_4 \text{Penny}_i \times \text{Post}_t + \beta_5 \text{ETF}_i \times \text{Post}_t + \delta_t + \gamma_i + \varepsilon_{i,t}.$$

We estimate both a fixed-effects linear probability model and fixed effects logit on a dichotomous variable equal to one if the option series has an order cancellation rate above the 75th percentile (mean order cancellation rate = 97.72%), and zero if the order cancellation rate is below the 25th percentile (mean order cancellation rate = 26.11%). We restrict the analysis to observations on the PHLX. *Post* is a dummy variable equal to one if the observation occurs after the fee change and zero otherwise. *Call* equals one if the option is a call and zero for a put. *Days-to-Expiration* is the number of days to option expiration. *In-the-Money* equals one if the option series is in-the-money and zero for out-of-the-money. *Penny* is a categorical variable equal to one if the option trades in pennies and zero otherwise. *ETF* equals one if the underlying security is an ETF and zero for a stock. *** denotes significance at the 0.01 level.

	(1)	(2)	
	LPM	Logit	Odds Ratio
Call x Post	-0.0066 (-0.2480)	-0.3385** (-2.0601)	0.7129
Days-to-Expiration x Post	0.0004 (1.7872)	0.0030*** (2.9480)	1.0030
In-the-Money x Post	0.0689*** (4.7316)	0.5949*** (7.8537)	1.8128
Penny x Post	-0.1124*** (-2.8711)	-1.1610*** (-4.3147)	0.3132
ETF x Post	0.1234*** (3.7397)	0.7224*** (3.0384)	2.0593
Constant	0.6944*** (40.9368)		
Day FE	Yes	Yes	
Option FE	Yes	Yes	
R ²	0.0753		
χ ²		1115.9507	
N	21,638	21,638	

Table D.1. Order Cancellation Fees, Order Flow, and Execution Quality – ETF Subsample

The table reports the results of estimating the following regression equation for the subsample of option series in exchange traded funds (ETFs) on average daily option series observations:

$$\begin{aligned} DepVar_{i,j,t} = & \alpha + \beta_1 Phlx_j + \beta_2 Phlx \times Post_{j,t} + \beta_3 Days - to - Expiration_{i,j,t} + \beta_4 S/X_{i,j,t} + \\ & \beta_5 Trade\ Price_{i,j,t} + \beta_6 Underlying\ MCAP_{i,j,t} + \beta_7 Log(Underlying\ Volume)_{i,j,t} + \\ & \delta_t + \gamma_i + \varepsilon_{i,j,t} . \end{aligned}$$

The dependent variable is set to one of the following: *Order Cancellation Rate*, *Order Duration*, *Log (# of Non-Marketable Orders)*, *Log (# of Marketable Orders)*, *Order Fill Rate*, *Marketable Order Fill Speed*, *Non-Marketable Order Fill Speed*, *Dollar Effective Spread*, *% Price Impact*, *Midpoint Volatility*, or *Range Volatility*. The variables have previously been defined. We include both day fixed effects and option series fixed effects. We report *t*-statistics in parentheses obtained from standard errors two-way clustered by option series and day. ***, **, denote statistical significance at the 0.01 and 0.05 levels, respectively.

Dependent Variable	Phlx × Post	t-stat	R ²	N
<i>Panel A. Order Flow</i>				
Order Cancellation Rate	-0.1171***	(-6.6938)	0.5805	17,444
Order Duration	272.6782***	(5.4176)	0.1502	17,237
log (# of Non-Marketable Orders)	-0.9664***	(-6.5129)	0.8279	17,444
log (# of Marketable Orders)	-0.1776**	(-2.1297)	0.0638	7,844
<i>Panel B. Order Execution Quality</i>				
Non-Marketable Order Fill Rate	0.0747***	(6.1928)	0.4674	17,444
Marketable Order Fill Speed	0.0021	(0.3533)	0.0680	7,844
Non-Marketable Order Fill Speed	48.1364	(0.7628)	0.0325	16,705
\$ Effective Spread	0.0109***	(3.1463)	0.0159	49,755
<i>Panel C. Pick-Off Risk</i>				
% Price Impact	-0.0307	(-0.8275)	0.0140	14,695
Midpoint Volatility	0.0689***	(4.2380)	0.0534	14,695
Range Volatility	0.1272***	(4.9146)	0.1158	14,695

Table E.1. Order Cancellation Fees, Order Flow, and Execution Quality – Bootstrap Analysis

The table reports the results of estimating the following regression equation using 1,000 bootstrap replications on average daily option series observations:

$$DepVar = \alpha + \beta_1 Phlx + \beta_2 Post + \beta_3 Phlx \times Post + \varepsilon.$$

The dependent variable is set to one of the following: *Order Cancellation Rate*, *Order Duration*, *Log (# of Non-Marketable Orders)*, *Log (# of Marketable Orders)*, *Order Fill Rate*, *Marketable Order Fill Speed*, *Non-Marketable Order Fill Speed*, *Dollar Effective Spread*, *% Price Impact*, *Midpoint Volatility*, or *Range Volatility*. The variables have previously been defined. To conserve space, we only report the interaction term between *Phlx* and *Post*. ***, **, denote statistical significance at the 0.01 and 0.05 levels, respectively.

Dependent Variable	Phlx × Post	t-stat	R ²	N
<i>Panel A. Order Flow</i>				
Order Cancellation Rate	-0.1197***	(-33.4520)	0.4793	65,760
Order Duration	217.4798***	(10.3325)	0.1077	64,971
log (# of Non-Marketable Orders)	-0.8246***	(-27.4315)	0.7348	65,760
log (# of Marketable Orders)	0.1309***	(3.8770)	0.0017	36,226
<i>Panel B. Order Execution Quality</i>				
Non-Marketable Order Fill Rate	0.0730***	-29.0019	0.3713	65,760
Marketable Order Fill Speed	-0.0101***	(-5.3481)	0.0257	36,226
Non-Marketable Order Fill Speed	3.0213	-0.0933	0.0076	61,851
\$ Effective Spread	0.0072***	(5.1422)	0.0011	186,335
<i>Panel C. Pick-Off Risk</i>				
% Price Impact	0.0678***	(3.4764)	0.0003	55,063
Midpoint Volatility	0.1174***	(14.5295)	0.0078	55,063
Range Volatility	0.0929***	(11.3151)	0.0037	55,063

Table E.2. Order Cancellation Fees, Order Flow, and Execution Quality – Bayesian Analysis

The table reports the results of estimating the following regression equation using Bayesian statistics. We use Gibbs sampling and 10,000 retained Markov chain Monte Carlo iterations on average daily option series observations:

$$DepVar = \alpha + \beta_1 Phlx + \beta_2 Post + \beta_3 Phlx \times Post + \varepsilon.$$

The dependent variable is set to one of the following: *Order Cancellation Rate*, *Order Duration*, *Log (# of Non-Marketable Orders)*, *Log (# of Marketable Orders)*, *Order Fill Rate*, *Marketable Order Fill Speed*, *Non-Marketable Order Fill Speed*, *Dollar Effective Spread*, *% Price Impact*, *Midpoint Volatility*, or *Range Volatility*. The variables have previously been defined. We report the estimates of posterior means, standard deviation, and Monte Carlo standard errors for the interaction term between *Phlx* and *Post*.

	Phlx × Post							
Dependent Variable	Mean	Std. Dev.	MCSE	Median	95% Cred. Interval		Efficiency Average	N
<i>Panel A. Order Flow</i>								
Order Cancellation Rate	-0.120	0.003	0.000	-0.120	-0.126	-0.113	1.000	65,760
Order Duration	225.552	19.453	0.197	225.479	187.300	264.475	0.974	64,971
log (# of Non-Marketable Orders)	-0.824	0.028	0.000	-0.824	-0.879	-0.770	0.996	65,760
log (# of Marketable Orders)	0.131	0.035	0.000	0.131	0.063	0.198	0.999	36,226
<i>Panel B. Order Execution Quality</i>								
Non-Marketable Order Fill Rate	0.073	0.002	0.000	0.073	0.068	0.078	1.000	65,760
Marketable Order Fill Speed	-0.010	0.002	0.000	-0.010	-0.014	-0.006	0.995	36,226
Non-Marketable Order Fill Speed	-3.797	29.447	0.298	-3.825	-60.960	53.221	0.995	61,851
\$ Effective Spread	0.007	0.002	0.000	0.007	0.003	0.011	1.000	186,335
<i>Panel C. Pick-Off Risk</i>								
% Price Impact	0.068	0.020	0.000	0.068	0.027	0.107	0.990	55,063
Midpoint Volatility	0.117	0.008	0.000	0.117	0.103	0.132	0.985	55,063
Range Volatility	0.093	0.008	0.000	0.093	0.077	0.108	0.984	55,063

Figure B.1. Order Cancellation Fees and Dollar Volume

Figure 5 plots average daily dollar volume for option series trading on the PHLX (dark solid line) and the NOM (light dotted line) around the event date, August 19th, 2010. *Dollar Volume* equals price times the number of contracts. We plot 10-day moving averages.

